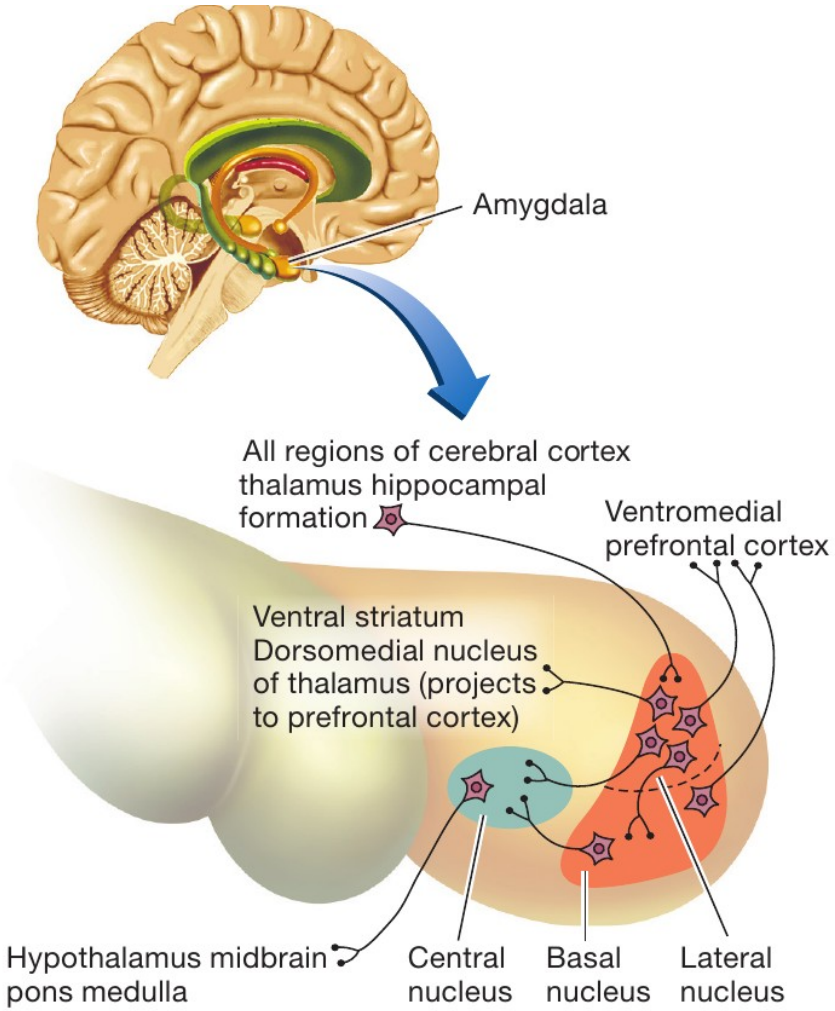


Carlson and Birkett (2022), Fig. 11.2, p. 340.

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Threat, Aggression, Regulation, Expression, and Bodily Feedback

Feeling and Acting: The Neural Systems of Emotion

Fall 2026

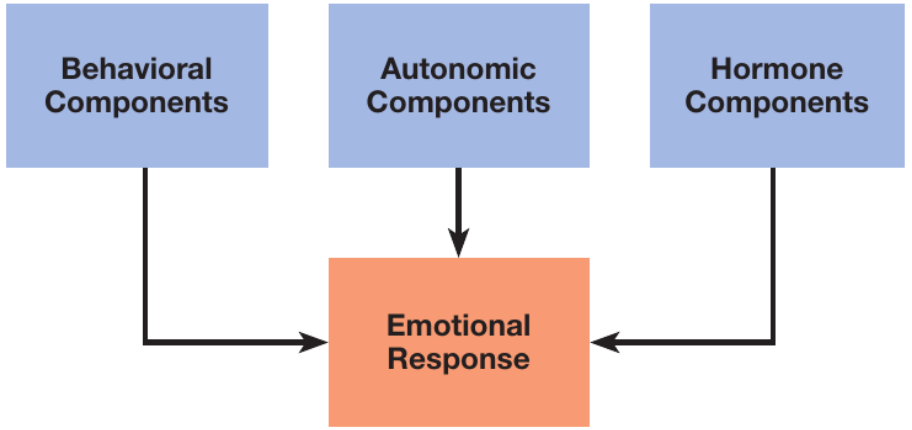
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

An emotional response has several measurable components

Emotion includes coordinated changes in behavior, bodily physiology, and subjective experience. A threatening event can produce withdrawal, altered breathing, and a reported feeling of fear. These components often interact, but measuring one does not automatically establish the presence or intensity of every other component in the same individual.

Autonomic responses regulate organs such as the heart, lungs, and digestive tract, while **endocrine responses** involve hormones released into the circulation. Behavioral responses organize actions, including approach, freezing, escape, and communication. Different neural pathways can therefore contribute to different parts of the response rather than producing one indivisible emotional output.

Subjective feeling is assessed through reports in people, whereas animal experiments usually measure actions, physiology, or neural activity. An observed defensive behavior provides evidence about a response system without granting direct access to the animal's private experience. Keeping these measurements distinct makes it possible to compare mechanisms across species without confusing the levels of explanation.



Carlson and Birkett (2022), Fig. 11.1, p. 339.



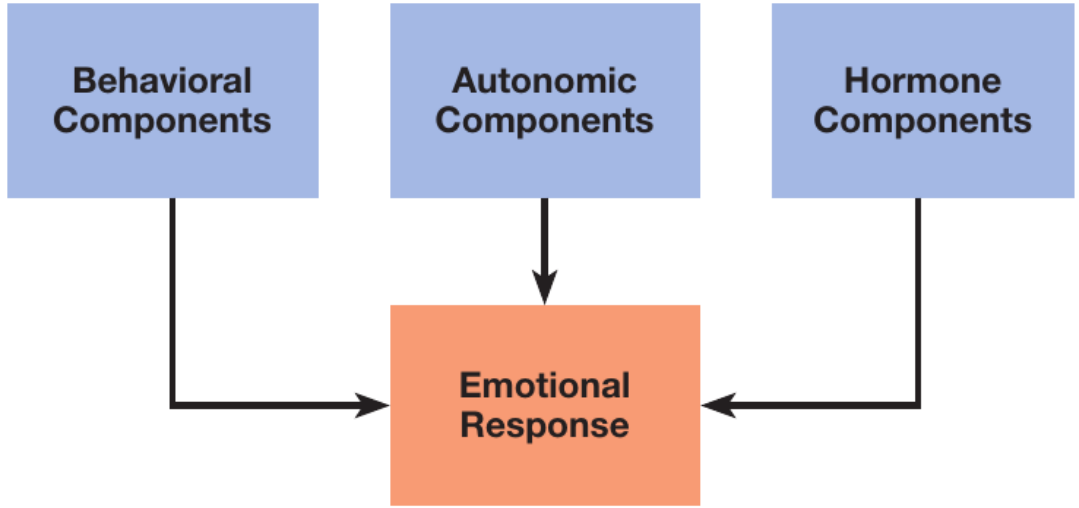
Kunwar et al. (2015), media7.

Valence differs from the importance of a stimulus

Valence describes whether an event is experienced or treated as positive or negative. **Salience** describes how strongly it captures processing because it matters in the current situation. A rewarding opportunity and a threat can both be highly salient even though their expected consequences have opposite valence.

Arousal concerns the level of physiological and behavioral activation. Arousal does not identify a unique emotion because excitement and fear can both increase it. Experiments therefore compare several kinds of events and several response measures before claiming that a neuron specifically represents unpleasantness, significance, or general activation.

The same physical stimulus can also change meaning with learning and context. A sound predicting food has a different consequence from the same sound predicting danger. Neural responses may track sensory properties, expected outcome, or the action required, so the interpretation depends on which of these variables the experiment changes independently.



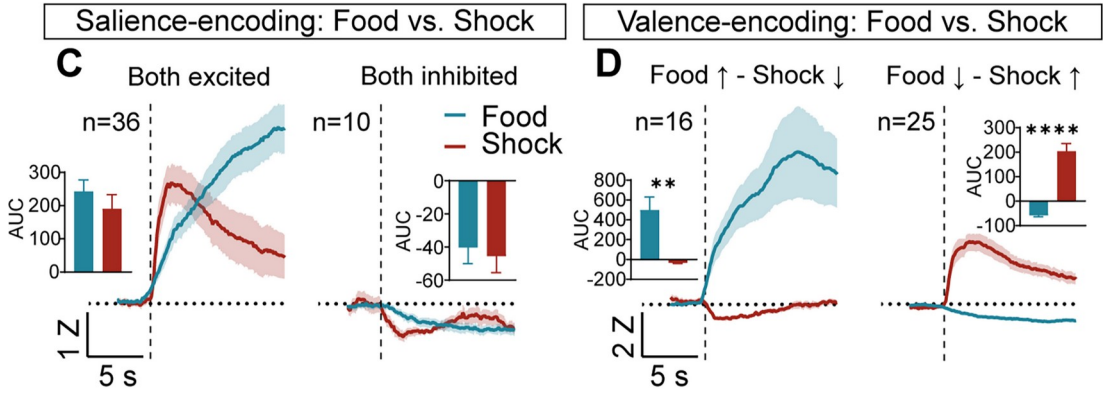
Carlson and Birkett (2022), Fig. 11.1, p. 339.

Amygdala cells distinguish positive and negative events

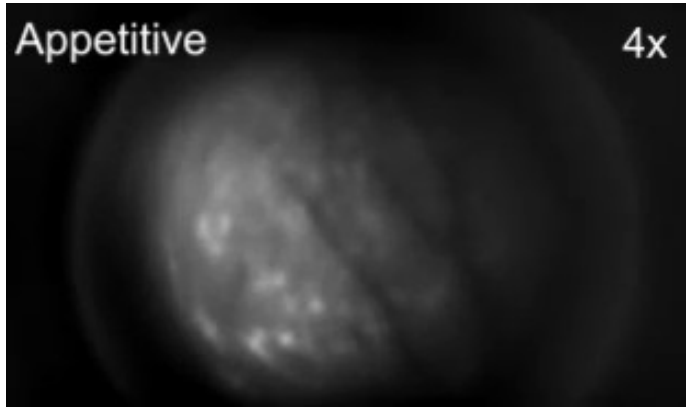
A mouse study recorded calcium activity from individual neurons in the **central amygdala**, or CeA, during appetitive and aversive conditioning. **Appetitive conditioning** establishes predictions about a desirable outcome. Comparing positive and negative outcomes allowed the investigators to test whether cells differentiated valence rather than responding to every important event equally.

Many recorded neurons showed different responses to oppositely valenced outcomes, while a smaller group responded in a manner more consistent with general salience. The central amygdala therefore contained heterogeneous representations. A response in this region could not be interpreted automatically as a dedicated measure of fear simply because the area participates in defensive behavior.

The same imaging approach followed the response patterns of identified cells across events. Calcium fluorescence reports a cellular activity-related signal with its own time course, rather than the exact timing of every spike. The comparison nevertheless separated neurons biased toward outcome valence from those broadly responsive to biologically important stimulation.



Kong et al. (2025), Fig. 5.



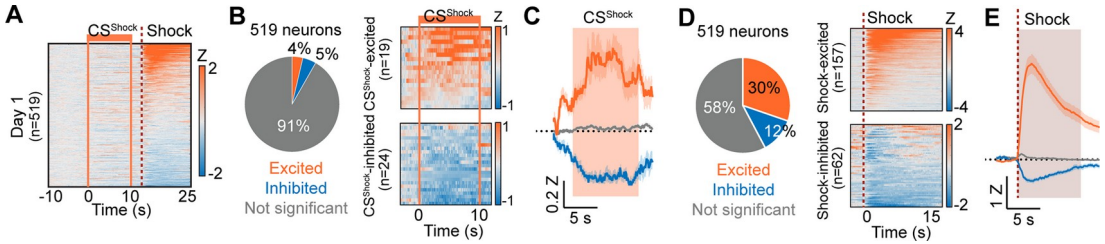
Kong et al. (2025), Video 1.

Learning changes amygdala responses to predictive cues

In the central-amygdala study, a **conditioned stimulus**, or CS, predicted an **unconditioned stimulus**, or US. The unconditioned event already had a biological consequence, while the cue gained predictive value through experience. Responses to the cue could therefore be compared with responses to the outcome it signaled.

Valence-related and salience-related cue representations were also detected, although they were less prominent than the corresponding outcome representations. A learned predictor did not simply reproduce every feature of the neural response to the event itself. The circuit distinguished information about what might happen from the actual delivery of that outcome.

This difference matters when interpreting a response to a warning signal. A cell can represent an expected event, the uncertainty surrounding it, or a learned action without directly reporting a subjective feeling. Tracking the same neurons during cue and outcome periods helps separate these possibilities within a region traditionally associated with emotional responses.



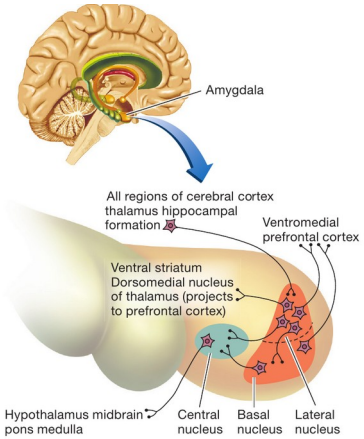
Kong et al. (2025), Fig. 4.

Amygdala nuclei have different inputs and outputs

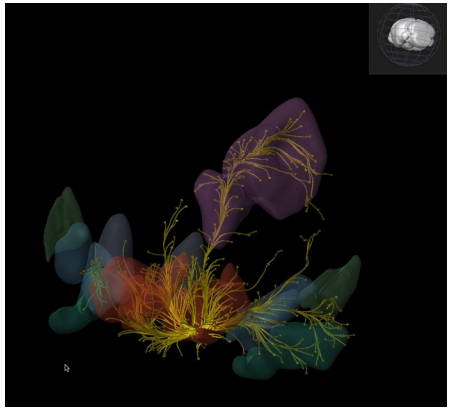
The **amygdala** is a collection of interconnected nuclei, groups of neurons with different cellular organization and connections. Lateral and basal regions receive substantial sensory and contextual information. Their interactions with cortical and hippocampal systems help relate current events to previous experience rather than processing every stimulus in isolation.

The central amygdala provides important routes to hypothalamic and brainstem systems controlling behavior and physiology. Its outputs can influence defensive immobility, autonomic regulation, arousal, and endocrine responses through different targets. A change in one output pathway need not alter every component of an emotional response in exactly the same way.

These pathways also receive feedback and interact with local inhibitory networks. The amygdala should therefore not be imagined as a single passive alarm that sends an identical command whenever a threatening image appears. Its contribution depends on the particular nucleus, neuronal population, input pattern, and downstream circuit being studied.



Carlson and Birkett (2022), Fig. 11.2, p. 340.



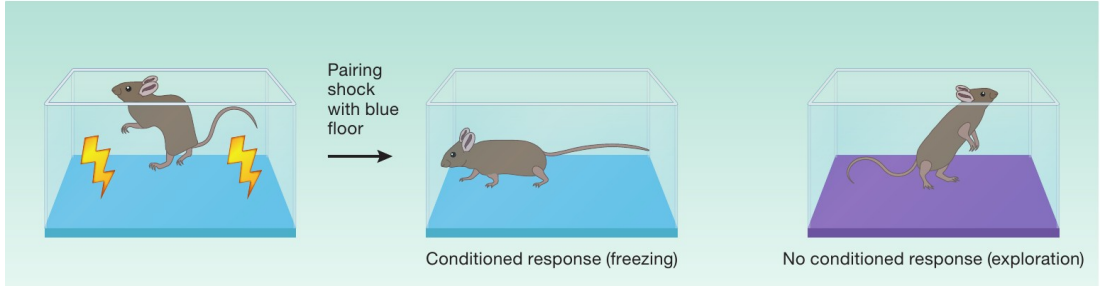
Kunwar et al. (2015), media9.

A neutral cue can acquire a defensive response

In **Pavlovian threat conditioning**, an initially neutral cue is paired with an aversive event. After learning, the cue can trigger defensive responses before that event occurs. The animal has acquired a prediction, allowing behavior and bodily regulation to change in advance of an expected consequence.

Researchers may measure freezing, heart rate, startle, pupil responses, or skin conductance. These measures describe different outputs and may not change identically. **Freezing** is a reduction in movement that can serve a defensive function; it should not be treated as a direct numerical readout of every aspect of fear.

Cue learning can also depend on the surrounding environment, the time between events, and previous experiences. A sound paired with an aversive event in one place need not produce the same response everywhere. Sensory input and contextual information jointly influence the circuits that select the learned defensive action.



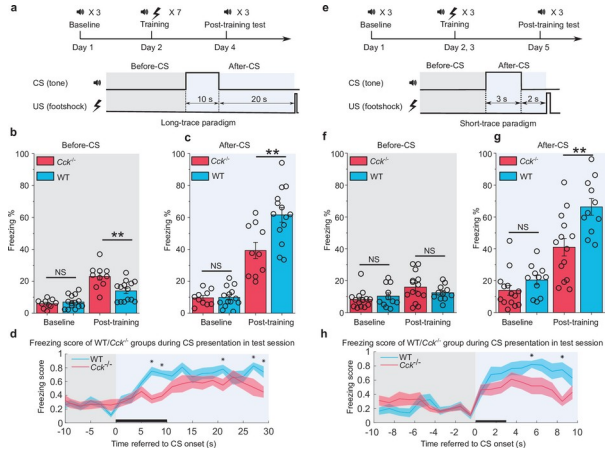
Carlson and Birkett (2022), Fig. 11.4, p. 341.

Trace conditioning requires bridging a gap between events

Trace conditioning inserts a gap between the end of a cue and the beginning of an aversive event. This requires the nervous system to retain information across time. A mouse study examined whether entorhinal-cortex neurons producing **cholecystokinin**, or CCK, contributed to learning across that interval.

Mice lacking normal CCK signaling showed reduced conditioned freezing after trace training. The researchers combined behavioral assays with recordings and pathway manipulations involving the lateral amygdala. The result linked a defined peptide-related pathway to a particular temporal learning problem rather than assuming all conditioning procedures depended on identical neural mechanisms.

The study also compared trace and delay arrangements, in which the cue and outcome have different temporal relationships. This comparison helped distinguish bridging an interval from merely hearing a cue or producing a movement response. A deficit in trace learning can therefore reveal altered temporal association without establishing a universal inability to detect danger.



Feng et al. (2021), Fig. 1.



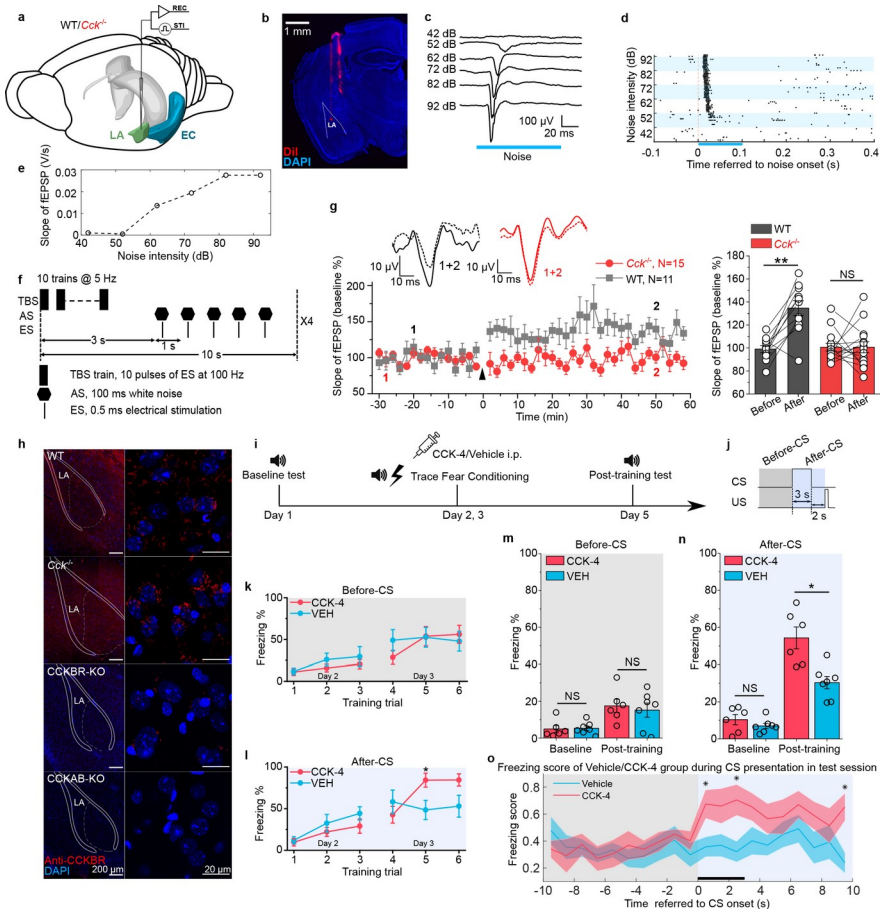
Feng et al. (2021), Video 1.

Peptide signaling enables plasticity during threat learning

The trace-conditioning study examined how CCK influences activity-dependent changes in the lateral amygdala. **Synaptic plasticity** is a lasting change in how effectively one neuron influences another. A peptide can regulate whether coincident activity produces that change, rather than carrying the entire sensory representation of the conditioned sound.

Pathway inhibition and peptide-related manipulations reduced learning-associated responses, while appropriate peptide treatment restored aspects of conditioning in the deficient mice. Comparing these conditions distinguished missing peptide support from an irreversible absence of every neuron needed for the learned behavior. The timing and target of the intervention helped identify the affected process.

Behavioral recordings were interpreted alongside neural and synaptic measurements. A reduced freezing response alone could have several causes, but a coordinated change in pathway activity, plasticity, and learning provides a more specific account. The proposed mechanism concerns how events become associated across a gap rather than how every emotional response is generated.



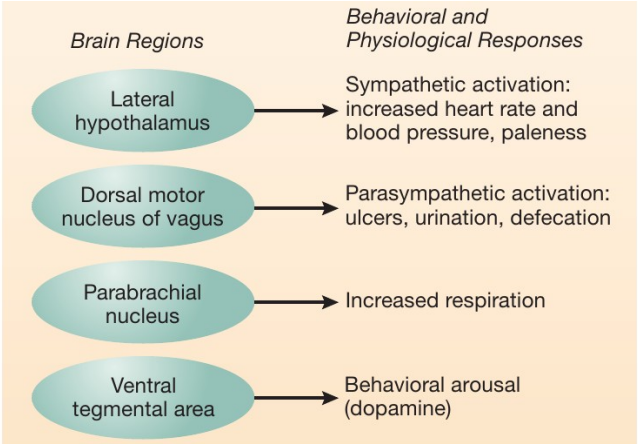
Feng et al. (2021), Fig. 2.

Brainstem outputs organize different parts of defense

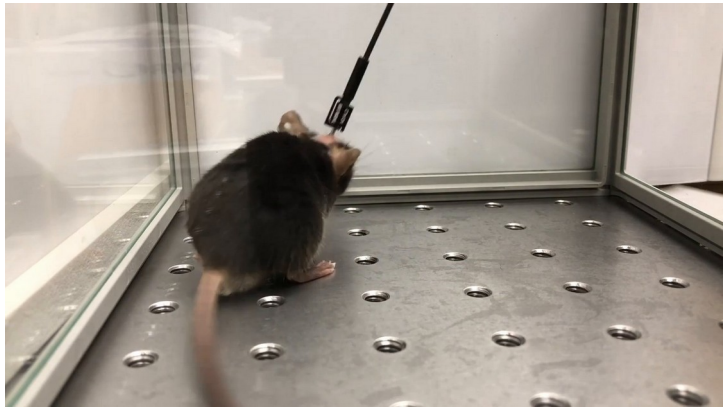
The **periaqueductal gray**, or PAG, surrounds the cerebral aqueduct in the midbrain and contains populations involved in defensive actions. Other brainstem and hypothalamic targets regulate breathing, circulation, arousal, and hormone secretion. These outputs help translate an assessed threat into a coordinated change in the state of the body.

Different PAG populations contribute to different actions, including immobility and escape. Their effects depend on input patterns and cell identity rather than a universal command from the entire structure. Defensive behavior also changes as a threat approaches, so the response appropriate at a distance may differ from the one selected during immediate contact.

Autonomic and motor components can be manipulated separately or together. An intervention that reduces movement does not necessarily produce the same cardiac or respiratory pattern as natural freezing. Measuring several outputs helps determine which parts of the defensive response were recruited by the particular circuit manipulation.



Carlson and Birkett (2022), Fig. 11.3, selected output branches, p. 340.



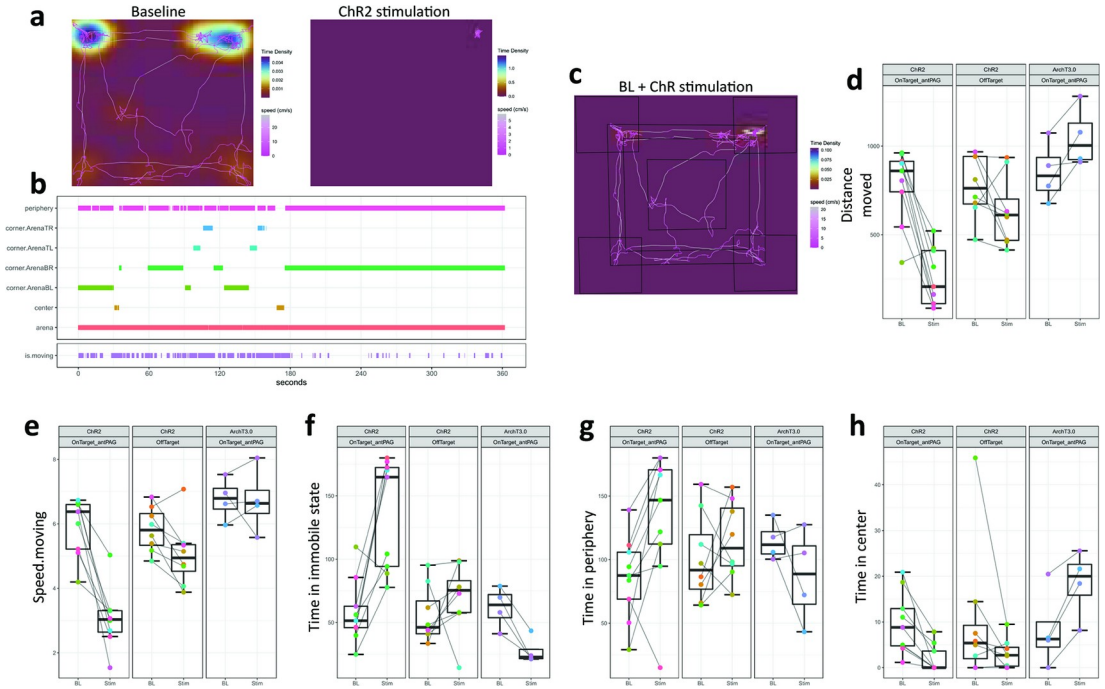
Vaaga et al. (2020), Video 1.

A hypothalamic projection can produce immobility

A mouse study targeted neurons marked by the developmental regulator **Foxb1** and their projections to the PAG. The researchers used light-sensitive channels to activate axon endings in a defined midbrain region. This tested an identified hypothalamic output route rather than stimulating every hypothalamic neuron or every PAG cell simultaneously.

Activating the projection produced abrupt immobility under the tested conditions. Some animals retained head movements while their limbs remained still, whereas others became more completely immobilized. The observed behavior therefore had identifiable components that could differ across animals rather than a single undifferentiated loss of every motor response.

The experiment established what activity imposed on this pathway could produce. It did not by itself prove that every natural instance of defensive freezing depends exclusively on that route. Combining the behavioral effect with anatomical targeting and physiological measurements helped identify a circuit capable of coordinating parts of a defensive-like response.



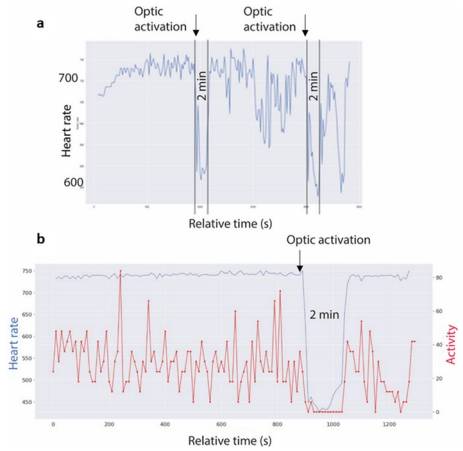
Cola et al. (2024), Fig. 3.

Respiration and heart rate need their own measurements

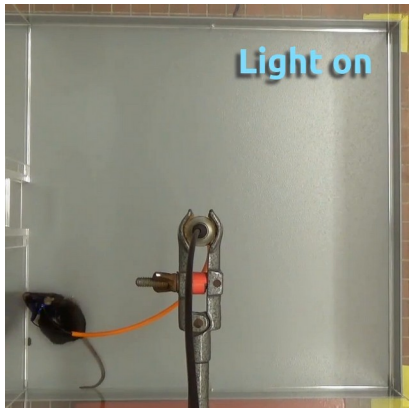
The Foxb1 study measured bodily responses as well as visible immobility. **Tachypnea** means increased breathing rate, while **bradycardia** means decreased heart rate. These changes can occur together because respiratory and cardiac control involve partly different pathways; general arousal does not require every organ to accelerate in parallel.

Chemogenetic activation of the targeted hypothalamic population altered respiratory parameters, while timed optical stimulation of its PAG endings produced immobility and cardiac slowing. The manipulations differed in location and timescale. Their outcomes should therefore be assigned to the actual intervention rather than assuming cell-body and terminal stimulation were identical experiments.

Whole-body respiratory recordings and telemetric heart measurements revealed physiological effects that a movement-only video could not establish. An animal that appears still can remain physiologically active. Breathing and circulation can be reorganized through separate autonomic routes, allowing increased respiratory activity to coexist with cardiac slowing rather than requiring a uniform reduction across the body.



Cola et al. (2024), Fig. 6.



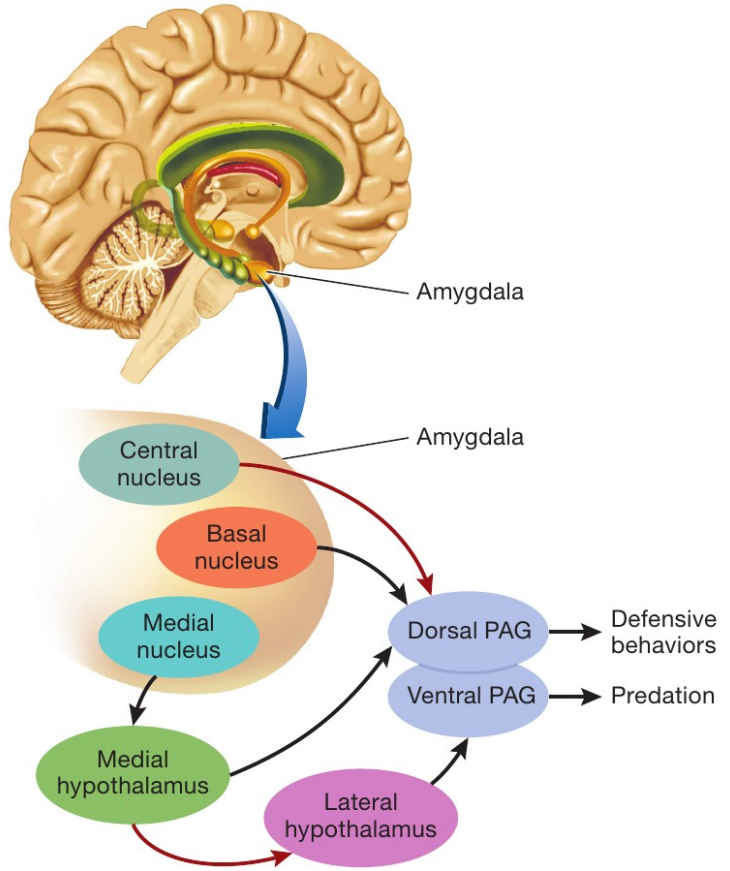
Cola et al. (2024), Video 2.

Defensive action depends on context and opportunity

An approaching threat can elicit investigation, cautious immobility, withdrawal, or rapid escape. The selected action depends on the threat's location and the availability of a safer place. **Action selection** means organizing one response while reducing competing alternatives, not merely increasing or decreasing a general quantity called fear.

Hypothalamic, amygdalar, cortical, and brainstem pathways contribute different information to this selection. Learned context can change how an ambiguous event is interpreted, while current body position constrains which movements are possible. A circuit that promotes escape may consequently have a different effect when no escape route is available.

Experiments distinguish these processes by measuring trajectories, speed, time near a threat, and periods of immobility. These endpoints can move in different directions after a manipulation. Their combination gives a more specific account of the animal's defensive strategy than treating every reduction in movement or approach as the same emotional change.



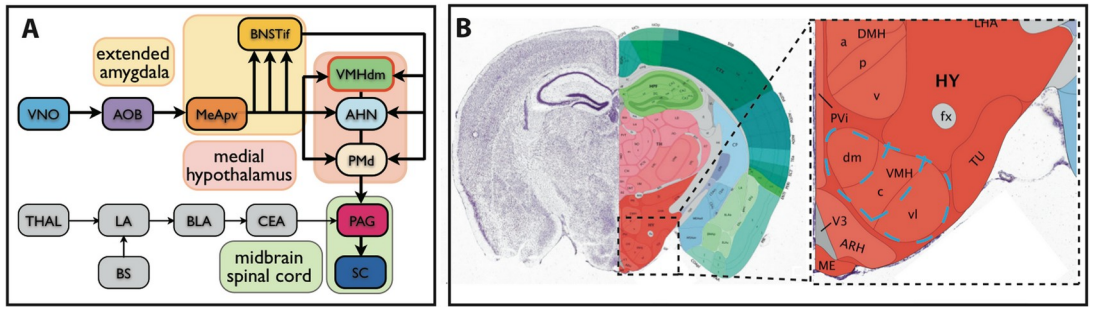
Carlson and Birkett (2022), Fig. 11.7, p. 345.

Hypothalamic activity can create a lasting defensive state

A mouse study targeted **SF1-expressing neurons** in the ventromedial hypothalamus. Selective optical activation produced defensive actions, including freezing and withdrawal. The investigators also examined whether effects persisted beyond stimulation and influenced subsequent behavior, asking whether the population altered an internal state rather than only triggering a brief muscle response.

Some defensive effects continued after the light ended and generalized beyond one narrowly specified action. The stimulation could also make a location aversive in choice tests. These findings indicated a change in how the animal responded to its environment rather than a motor reflex tied exclusively to the presence of light.

The experiment used behavioral properties such as persistence, negative valence, and generalization to characterize the induced state. These properties provide a mechanistic framework for comparing state-like responses across species. They do not require assuming that the animal's subjective experience is identical to a particular human description of fear.



Kunwar et al. (2015), Fig. 1.



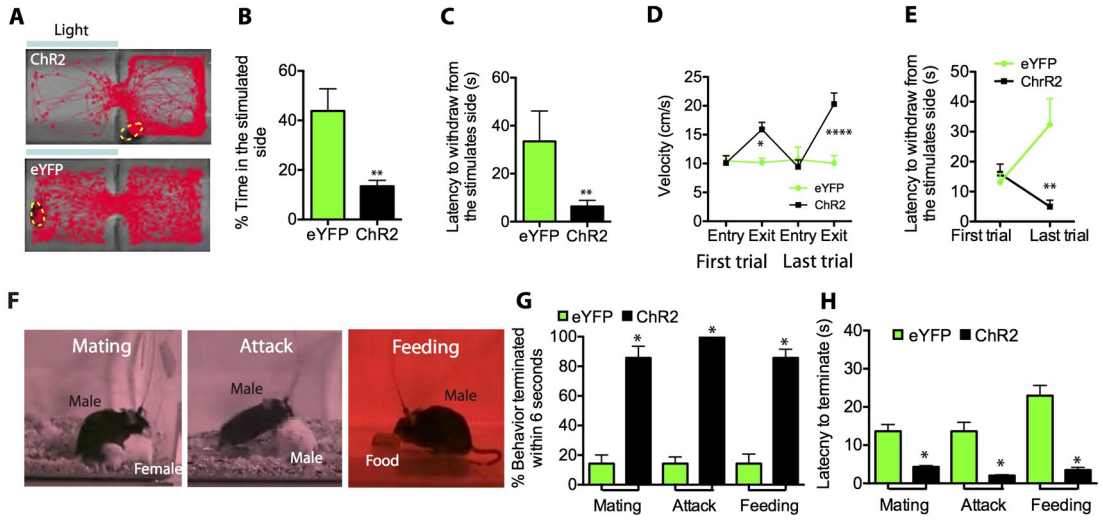
Kunwar et al. (2015), media1.

Activation strength changes the defensive action

The ventromedial-hypothalamus study varied stimulation conditions and examined the resulting actions. **Scalability** means that changing the strength of a manipulation changes a response in an ordered way. Increasing circuit drive could shift behavior from immobility toward more vigorous defensive movement rather than merely producing more of one fixed action.

The investigators also tested whether manipulating the population affected responses in naturally threatening settings. These tests addressed a different question from whether artificial activation was sufficient to produce a behavior. Necessity and sufficiency experiments together help distinguish a circuit's normal contribution from actions it can generate when driven experimentally.

Location avoidance and post-stimulation behavior added information beyond the movement observed during stimulation. An internal defensive state can bias later decisions after the immediate trigger has disappeared. This temporal extension helps explain why a brief neural event can influence several successive actions rather than only one moment of immobility.



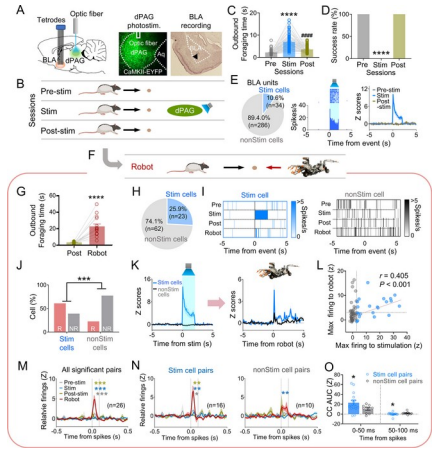
Kunwar et al. (2015), Fig. 4.

Midbrain defense signals can influence amygdala activity

A rat study recorded neurons while animals approached food in a setting containing a rapidly advancing robot predator. The **dorsal PAG** and **basolateral amygdala**, or BLA, were examined as interacting systems. The task combined the motivation to obtain food with the need to avoid a threat rather than presenting only an isolated warning tone.

Optical stimulation of the dorsal PAG caused fleeing and increased activity in subsets of BLA neurons. Cells responsive to that stimulation also responded strongly to the robot threat. The result showed that communication could influence amygdala processing from a midbrain defensive system, rather than flowing only from amygdala to downstream motor centers.

Anatomical tracing suggested possible indirect routes through midline thalamic structures. That anatomy does not establish that every recorded response traveled through one direct PAG-to-amygdala synapse. The combined experiments identified interacting populations during naturalistic avoidance while keeping the demonstrated activity changes separate from the proposed intervening pathway.



Kim et al. (2024), Fig. 3.



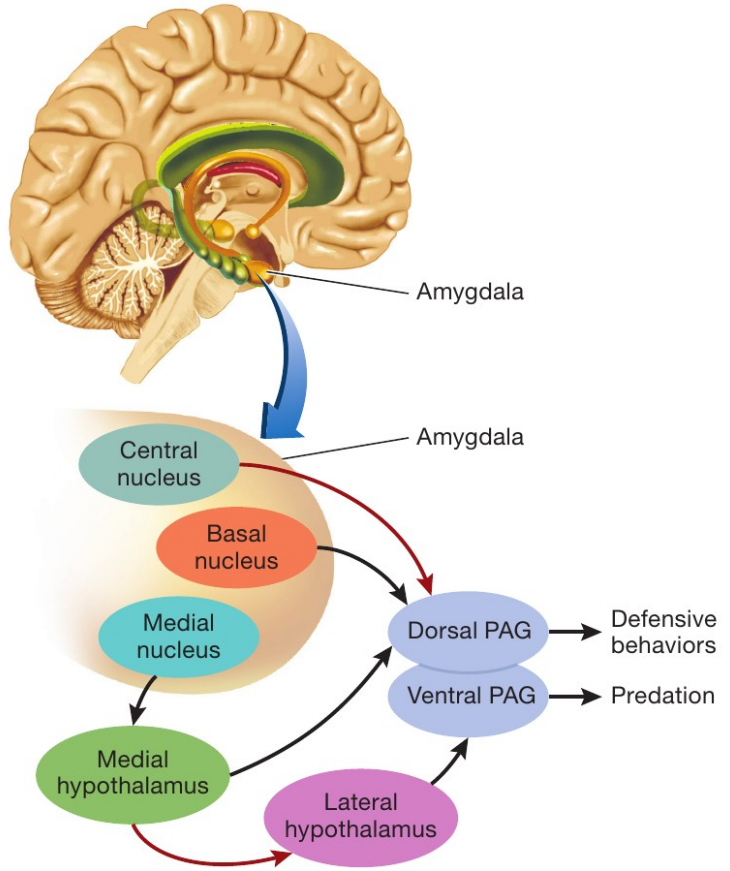
Kunwar et al. (2015), media2.

Defense and predation recruit partly different organizations

Defensive behavior reduces exposure to a threat, whereas **predation** obtains another animal as food. Both can involve rapid pursuit, contact, and biting, so similar movements do not establish identical motivation or circuitry. The relevant stimulus, physiological state, and outcome must be considered when classifying an aggressive-looking action.

Amygdalar and hypothalamic pathways communicate with different PAG and brainstem populations involved in these actions. Some routes bias avoidance or defensive attack, while others contribute to prey capture. Modern cell-specific results refine the older regional diagrams, showing that neighboring populations can have different or opposing behavioral effects.

An animal can also switch strategies within one encounter. Distance from the threat, prior defeat, hunger, and the presence of an escape route change the value of acting. Recording several components of behavior helps identify whether a manipulation changes threat assessment, the selected strategy, or execution of the final movement.



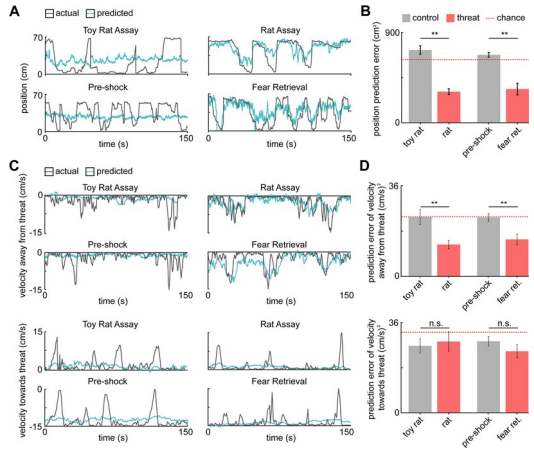
Carlson and Birkett (2022), Fig. 11.7, p. 345.

A hypothalamic–midbrain pathway regulates escape vigor

A mouse study investigated CCK-expressing neurons in the **dorsal premammillary nucleus**, a hypothalamic region, and their PAG projection. **Escape vigor** concerns how rapidly or strongly an animal executes an escape. The researchers distinguished the speed of fleeing from other defensive endpoints such as freezing or cautious investigation.

The targeted population was active around escape, and its ensemble activity contained information about movement and threat location. Inhibiting the population reduced escape speed during both innate and learned threat conditions. Projection inhibition also implicated the route to dorsolateral PAG, rather than treating every output of the hypothalamic region as equivalent.

The study separately examined related human imaging responses to aversive pictures. Those blood-oxygen signals suggested possible cross-species relevance but did not reproduce the causal mouse manipulation. The direct mechanistic result concerned how a defined mouse pathway contributed to the vigor of a defensive action once that action was being organized.



Wang et al. (2021), Fig. 4.



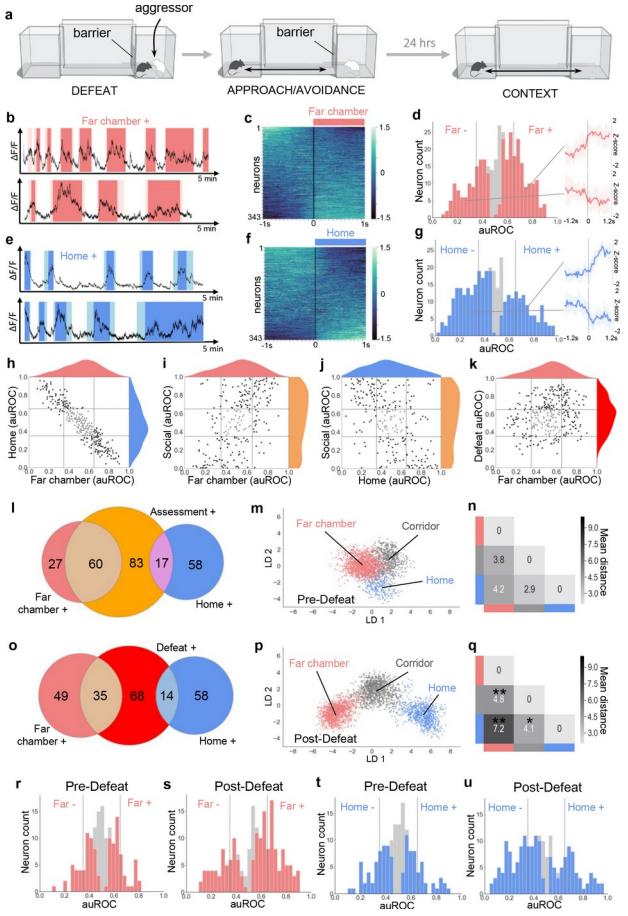
Kunwar et al. (2015), media5.

Social defeat changes what the hypothalamus represents

A mouse study followed neurons in ventromedial hypothalamus during social threat and after defeat. **Social defeat** occurs when an animal loses an aggressive encounter. The researchers asked whether the resulting neural response depended only on the opponent's presence or also incorporated information about the place where the threatening interaction occurred.

Cells responded both to social threat and to a chamber associated with previous defeat. When animals moved between home and defeat-related spaces, activity also represented their position within that arrangement. The hypothalamic signal therefore incorporated learned spatial context rather than remaining a fixed detector of a visible aggressor.

The experiment tracked neural activity during behavior, allowing changing responses to be related to the animal's own movement and experience. Contextual encoding does not mean that every cell contained a complete map. Different neurons contributed information about the threat, the surroundings, and the state in which the animal encountered them.



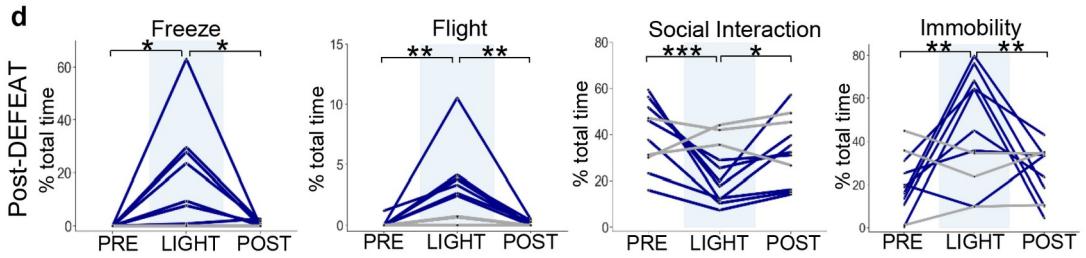
Krzywkowski et al. (2020), Fig. 2.

Experience changes the effect of activating a social circuit

The social-defeat study tested optical activation of **Esr1-expressing hypothalamic neurons** before and after defeat. Esr1 encodes estrogen receptor α, but the stimulation directly changed neuronal activity rather than administering estrogen. The genetic marker identified a population whose functional effect could be compared across different histories of social experience.

After defeat, activation could elicit avoidance that was not produced in the same way before defeat. The imposed drive therefore acted on a circuit whose behavioral output had been reorganized by experience. A stable molecular marker did not guarantee a fixed action every time the labeled neurons were stimulated.

The result separates cell identity from current network function. Synaptic inputs, downstream responsiveness, and learned context can change while a cell continues to express the same marker. Identifying an aggression-related population is therefore only part of the explanation; its effect also depends on the animal's experience and the circuit state at testing.



Krzywkowski et al. (2020), Fig. 5.



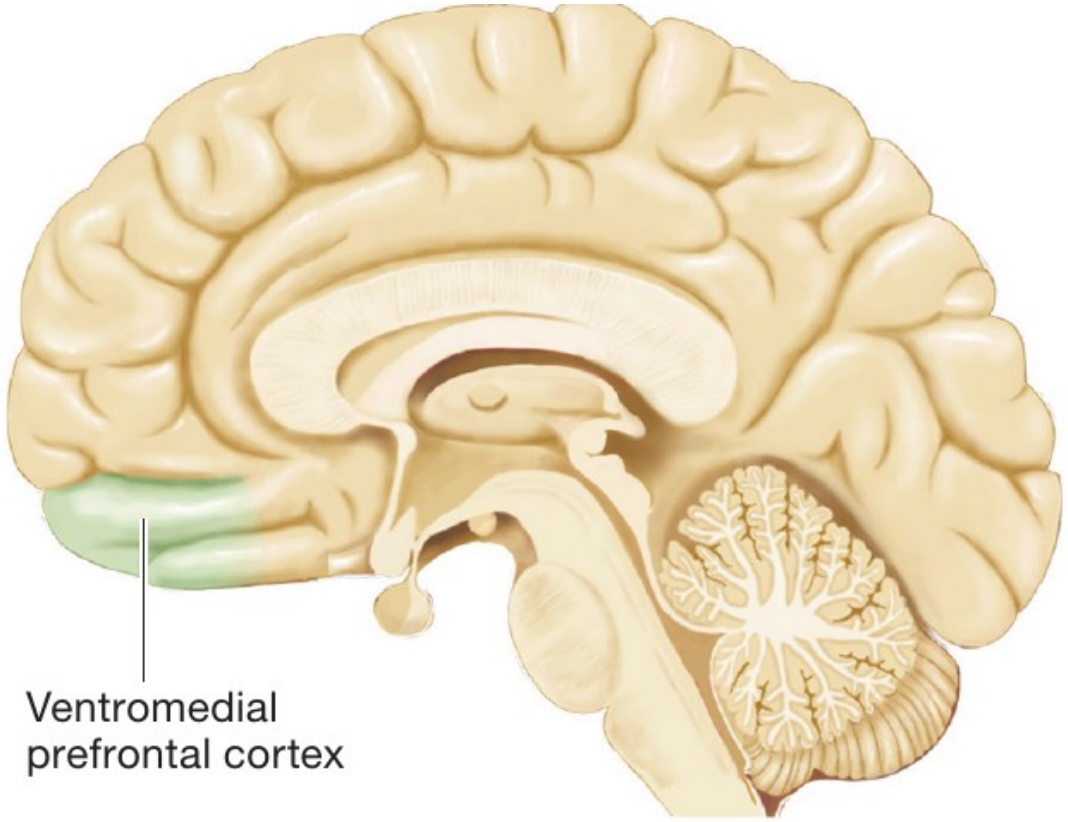
Krzywkowski et al. (2020), Video 3.

Anxiety differs from an immediate defensive response

Fear often refers to responding to a relatively identifiable and immediate threat, while **anxiety** often concerns uncertain or more distant danger. These are useful distinctions rather than perfectly isolated biological categories. Human reports, avoidance behavior, and physiological measurements can emphasize different features of the same developing situation.

Animal anxiety-related tasks often measure conflict between exploration and avoiding exposed spaces. Reduced time in an open area can reflect altered threat sensitivity, but also changes in movement, sensory function, or motivation. The interpretation depends on how the task and accompanying measurements separate those possibilities.

Neuromodulatory systems can change responsiveness across many potential events without specifying one concrete threat. For example, a change in norepinephrine input can alter processing in amygdalar circuits. Such a signal can bias attention and action under uncertainty while leaving the exact response dependent on the current sensory and spatial context.



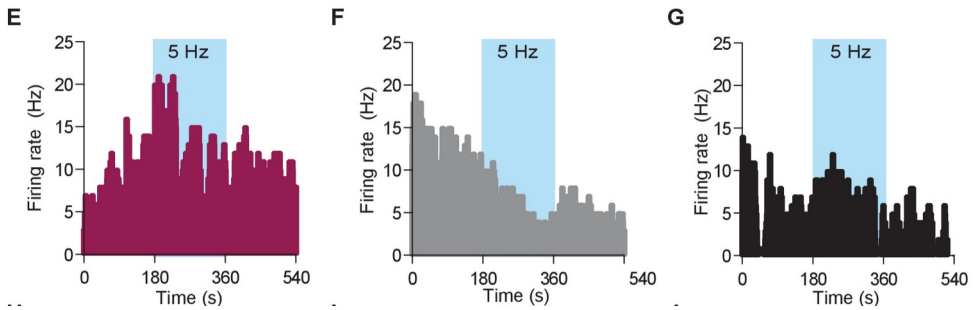
Carlson and Birkett (2022), Fig. 11.6, p. 342.

Norepinephrine input changes amygdala-related avoidance

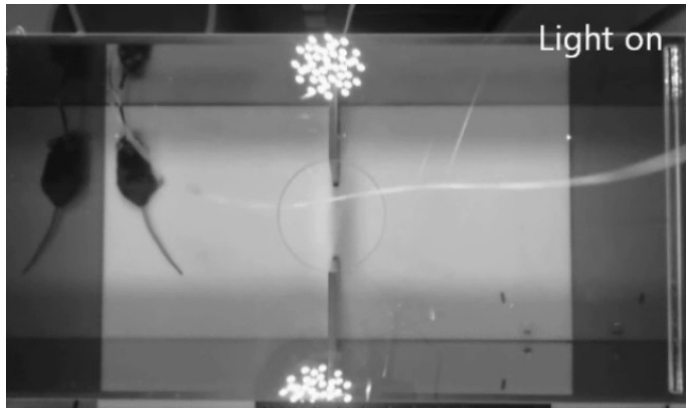
The **locus coeruleus** is a brainstem source of **norepinephrine**, a transmitter that regulates arousal and responsiveness. A mouse study activated its axon endings in the basolateral amygdala. This allowed the investigators to examine one output pathway rather than interpreting stimulation of the entire locus coeruleus as a single uniform change throughout the brain.

Terminal stimulation released norepinephrine, altered amygdala activity, and increased avoidance-related behavior in the tested tasks. It could also establish aversion to a stimulation-associated setting. A neuromodulatory projection therefore changed both local circuit activity and the way an animal allocated behavior in situations involving potential risk.

The experiment combined transmitter measurements, electrophysiology, anatomy, and behavior. Each method identified a different step between source neurons and action. The result concerned a defined norepinephrine projection and should not be generalized to mean that every increase in norepinephrine throughout the nervous system produces the same anxious response.



McCall et al. (2017), Fig. 3.



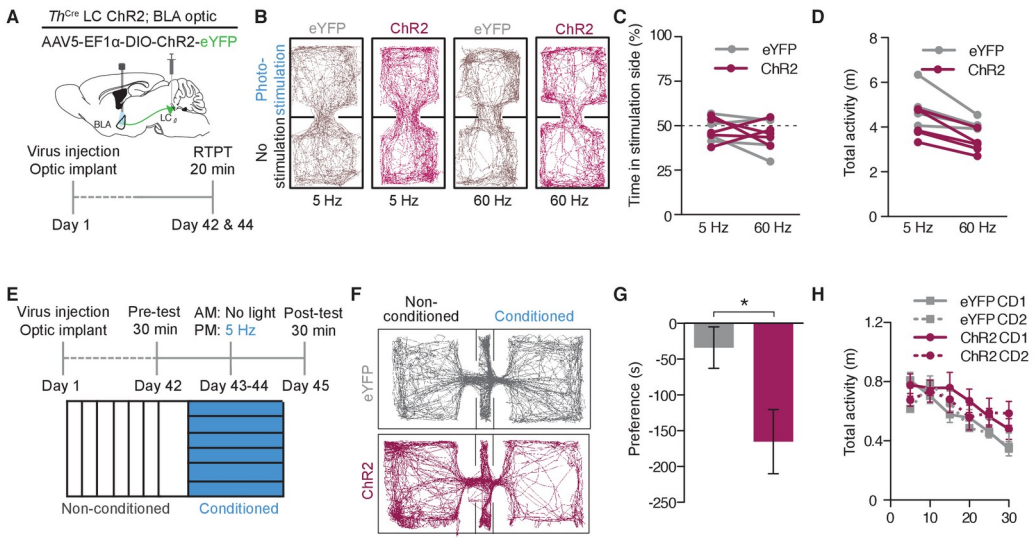
Kunwar et al. (2015), media3.

The receiving receptor determines a transmitter’s effect

Norepinephrine acts through several receptor families expressed by different target cells. In the locus-coeruleus study, local pharmacology tested **β-adrenergic receptors** in the basolateral amygdala. Blocking a receptor asks whether the transmitter’s effect at the target depends on that receiving mechanism rather than only on activity in the source neurons.

Local β-receptor blockade prevented the tested anxiety-related phenotype associated with enhanced norepinephrine input. The pathway therefore required an identified receptor contribution in the receiving region. Transmitter release alone did not determine the behavioral effect independently of the receptor machinery available to respond to it.

This organization explains how one broadly projecting transmitter system can have different functions at different targets. Receptor subtype, cell identity, and local connectivity change the consequence of the same chemical signal. A drug affecting norepinephrine cannot be understood completely by labeling the transmitter as simply calming or simply fear-producing.



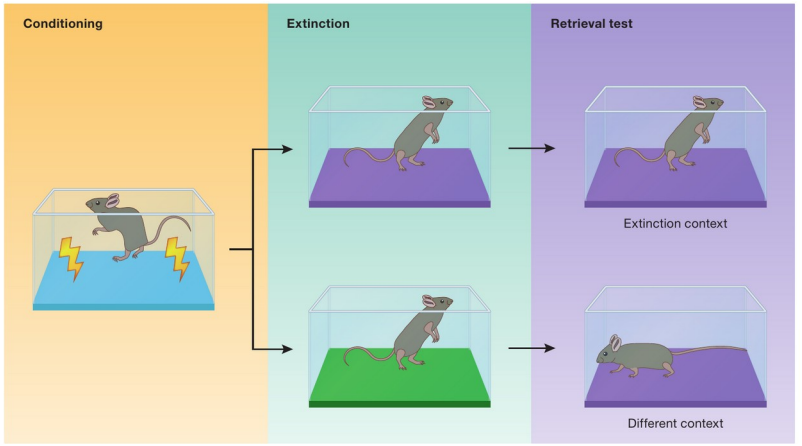
McCall et al. (2017), Fig. 5.

Extinction adds learning without erasing every association

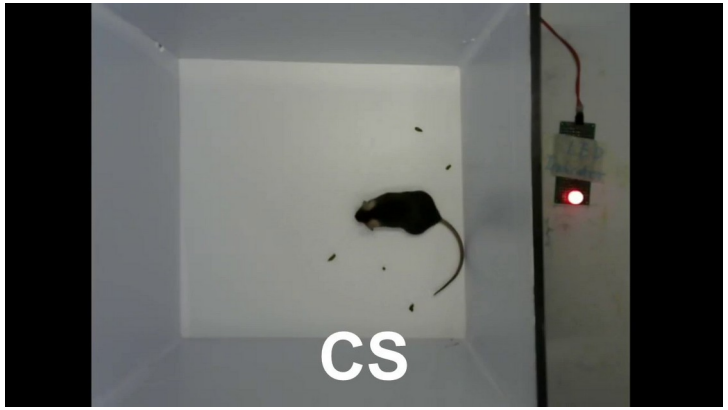
Extinction occurs when a conditioned cue is repeatedly presented without the expected aversive outcome. Defensive responding often declines, but this does not demonstrate that the original association has been physically erased. The animal may learn that the cue predicts a different outcome under the current conditions.

Renewal is the return of a conditioned response after a change of context, and **spontaneous recovery** is its return after time has passed. These observations show why a low response immediately after extinction is not equivalent to permanent loss of the original memory. Testing conditions influence which learned information controls behavior.

Prefrontal, hippocampal, amygdalar, and reward-related networks contribute to the learning and expression of these changes. A circuit manipulation can affect acquiring extinction, recalling it later, or selecting a defensive output. The timing of the manipulation is therefore needed to identify which stage of the process was altered.



Carlson and Birkett (2022), Fig. 11.5, p. 342.



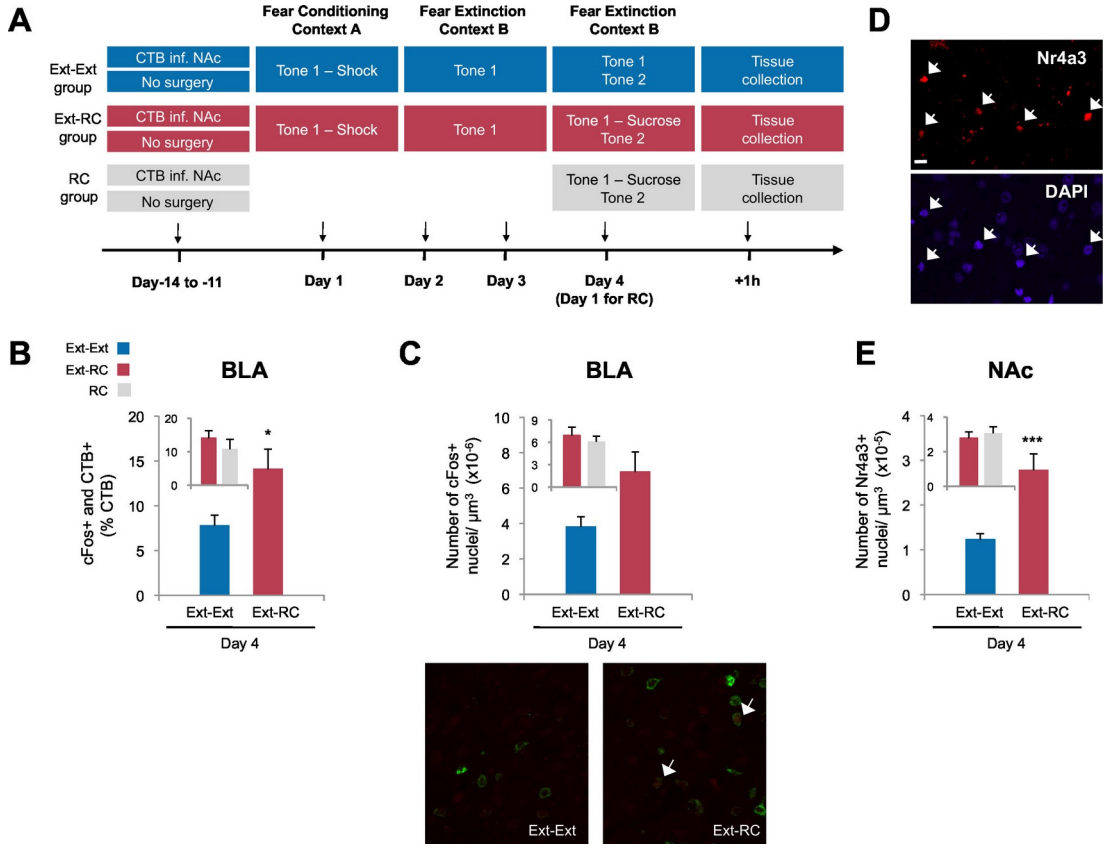
Feng et al. (2021), Video 3.

A reward-related amygdala output can strengthen extinction

A mouse study examined projections from basolateral amygdala to **nucleus accumbens**, a ventral-striatal region involved in motivated behavior. The investigators tested whether recruitment of this pathway during extinction affected the later return of defensive responses. They distinguished enduring extinction from a temporary reduction in freezing during the training session.

Pairing reward with extinction or optically stimulating the identified pathway during extinction reduced later return of conditioned responding. The intervention therefore influenced how persistent the new response pattern became. Recruitment of an appetitive circuit could alter an aversive memory process without requiring removal of every neuron representing the original threat cue.

The timing of stimulation tied the manipulation to new learning about the conditioned cue. Stimulating a circuit during extinction is different from merely suppressing movement during a later test. The result identified a pathway through which reward-related processing could improve the persistence of a learned change in defensive behavior.



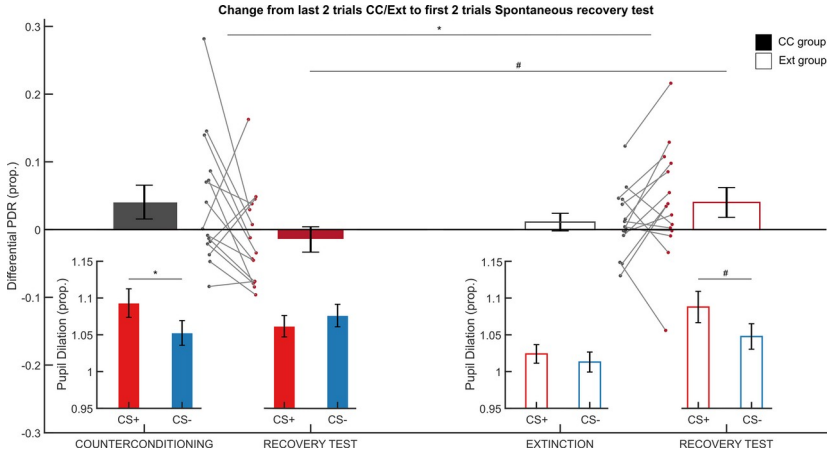
Correia et al. (2016), Fig. 3.

Counterconditioning gives a warning cue a new outcome

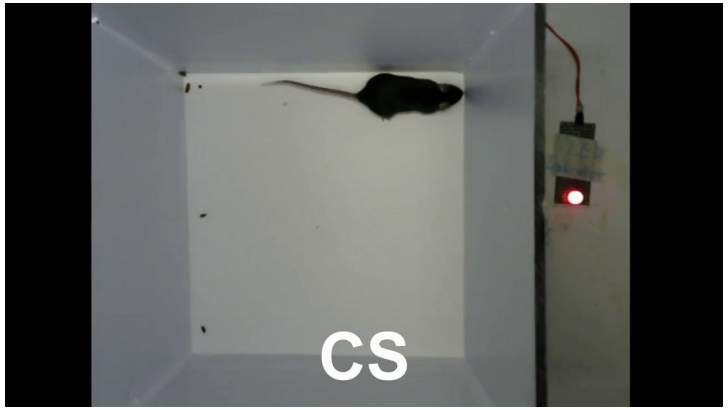
Counterconditioning pairs a previously aversive cue with an outcome of the opposite valence, such as a reward. A human experiment compared this procedure with ordinary extinction. Participants first learned threat associations involving picture categories and then experienced either reward-paired updating or cue presentations without the original aversive outcome.

The following day, physiological threat responses returned in the extinction condition but were reduced more persistently after counterconditioning under the tested conditions. The researchers measured pupil responses and explicit judgments rather than assuming that one physiological change represented every aspect of participants' subjective fear.

Imaging showed different patterns of activity during the two updating procedures, including recruitment of nucleus accumbens during counterconditioning. The procedure did more than remove the expected aversive event: it established a new appetitive consequence. This distinction helps explain why adding an opposing outcome can differ from simply presenting a cue safely.



Wirz et al. (2026), Fig. 3.



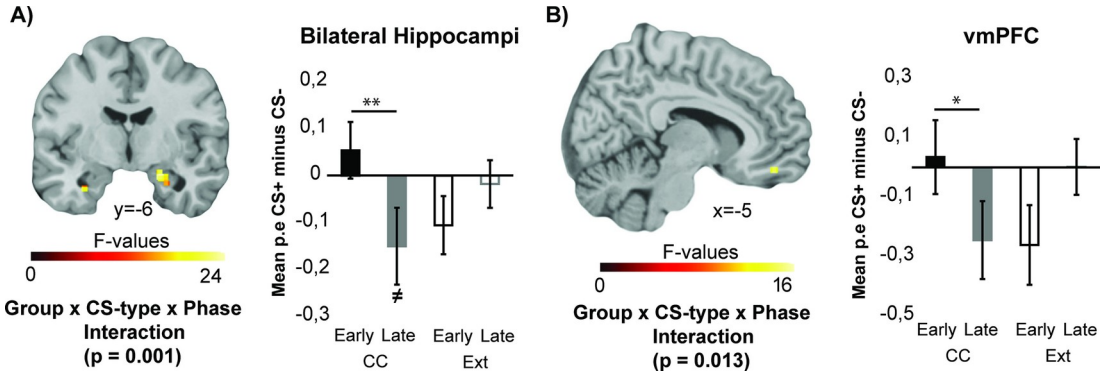
Feng et al. (2021), Video 4.

New associations can change memory for earlier events

The human counterconditioning study tested recognition of individual pictures after the learning procedures. **Episodic recognition** concerns remembering a previously encountered item rather than only responding to its category. This allowed the investigators to ask whether changing a cue's expected consequence also altered memory for specific experiences associated with that cue.

Counterconditioning strengthened recognition for relevant exemplars presented during updating and also affected memory for earlier threat-learning items. A later learning experience could therefore influence retention of material encountered during an earlier phase. The result extended beyond the immediate pupil response to a new warning-cue presentation.

The imaging comparison also distinguished hippocampal and ventromedial-prefrontal activity during counterconditioning and extinction. These blood-oxygen responses do not identify every synaptic change underlying memory. They show that two procedures producing reduced defensive responses can engage distinguishable networks and have different consequences for the remembered events.



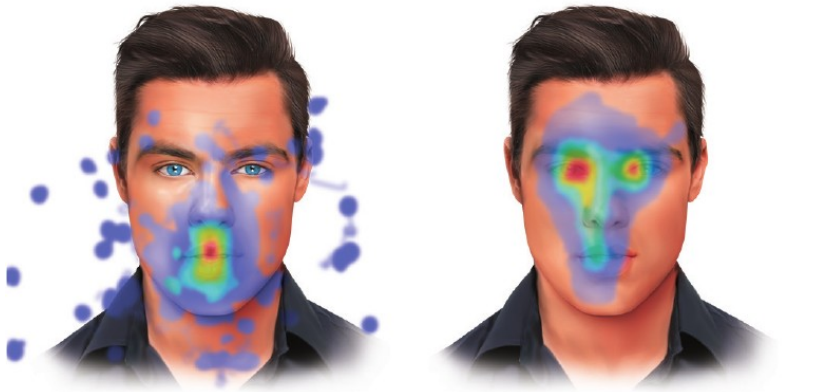
Wirz et al. (2026), Fig. 5.

Amygdala damage affects particular fear-related processes

Human lesion studies show that amygdala damage can alter threat learning, attention to social cues, and recognition of fearful expressions. **A lesion** is damage to tissue, and its behavioral effect depends on location, extent, timing, and the remaining network. A person with damage provides information about a functioning brain reorganized around that injury.

Some people with bilateral amygdala damage show unusually weak responses to particular external threats or difficulty extracting fear-related facial information. These findings identify important contributions of the amygdala. They do not establish that the person lacks every bodily or subjective response to every possible threatening condition.

Different threats can recruit different routes, including internal bodily signals and learned conceptual information. Subjective reports, recognition judgments, and physiological reactions must therefore be assessed separately. The absence of one response after a lesion cannot be converted into a claim that one brain region contains the entire human feeling of fear.



Carlson and Birkett (2022), Fig. 11.25, p. 363.



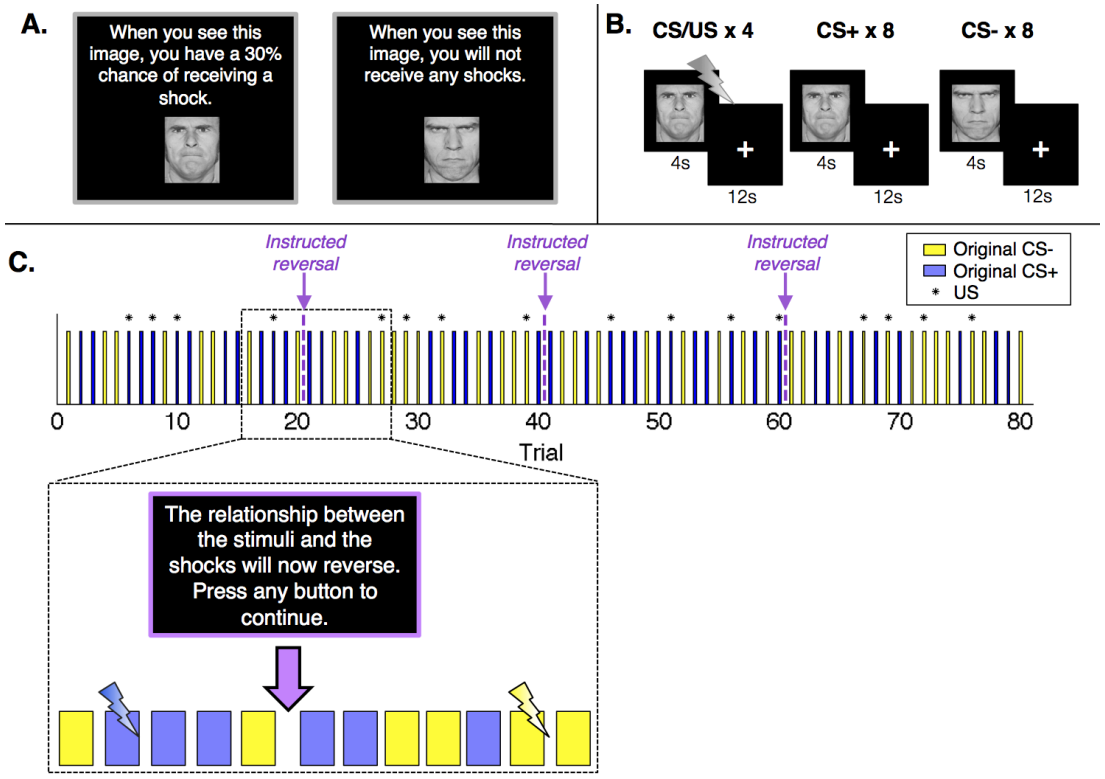
Kunwar et al. (2015), media8.

Instructions can update threat-learning responses

A human fMRI experiment compared learning from aversive feedback with explicit information about changing contingencies. **A contingency** is the relationship between a cue, an action, and an outcome. Instructions can tell a participant that this relationship has changed before repeated direct experience provides the same information.

Behavior and activity in striatum and orbitofrontal cortex updated with the instructed change. These responses incorporated knowledge conveyed through language rather than relying only on the most recent aversive outcome. A participant could therefore revise expectations through information that was not itself a new physical punishment.

The experiment compared informed and uninformed learning conditions to separate instruction-sensitive processing from feedback-driven updating. The result shows that threat-related behavior is not controlled solely by an unchanging reflex pathway. Higher-level knowledge can rapidly alter some components of prediction and choice in an otherwise similar aversive task.



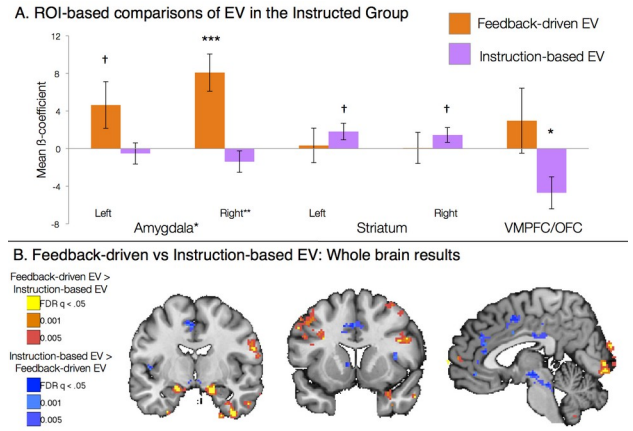
Atlas et al. (2016), Fig. 1.

Neural systems update differently after instructions

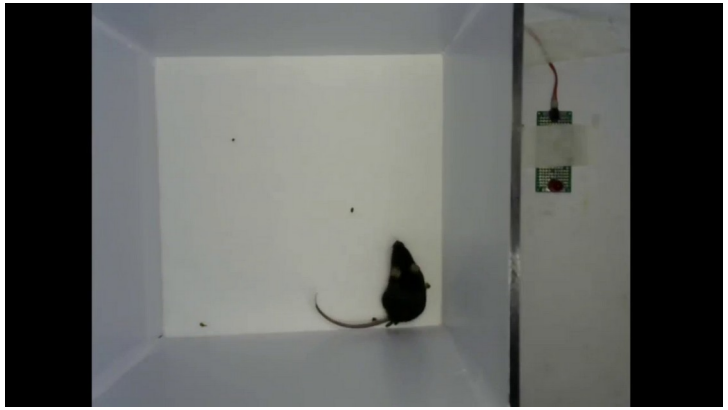
In the instructed-learning experiment, amygdala responses continued to reflect reinforcement history more strongly than the explicit reversal information. Other regions updated more immediately with instructions. The same participant therefore showed different forms of adaptation across measured neural systems rather than one brain-wide switch from an old association to a new one.

This dissociation helps explain why knowing that a situation is safe and changing every learned bodily response need not occur at the same pace. The experiment measured defined task responses, not a universal separation between rational thought and emotion. Several networks contributed information under different updating rules.

Imaging associations also do not mean that a region independently produces an entire feeling. The study compared sensitivity to instructions and feedback within a controlled learning problem. Its biological contribution was to distinguish pathways of updating, giving a more specific account than assigning all cognition to cortex and all emotion to amygdala.



Atlas et al. (2016), Fig. 4.



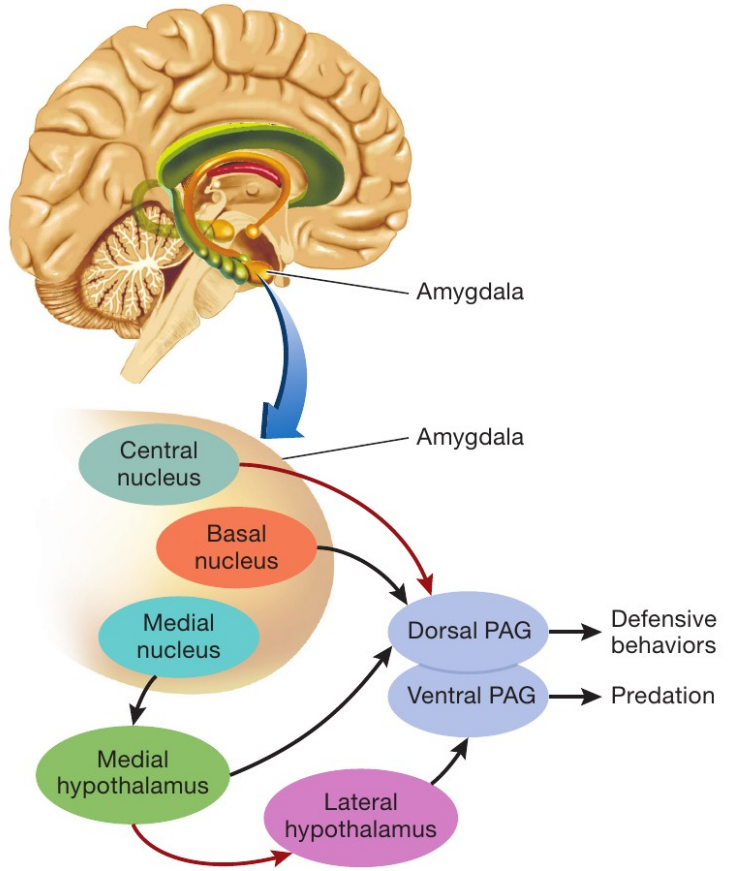
Feng et al. (2021), Video 5.

Aggression includes different actions and social functions

Aggression includes threatening displays and actions that can harm or displace another individual. Territorial defense, competition, defensive attack, and infant-directed aggression occur in different settings and can recruit different circuit components. Their visible movements may overlap without sharing the same trigger, motivation, or outcome.

Animal experiments often quantify approach, threat display, chase, attack, and withdrawal separately. A manipulation can increase contact without increasing biting, or reduce attack by changing sensory investigation. A single total score can conceal the stage of the encounter where the neural intervention actually acts.

Human violence involves additional psychological, developmental, and social processes. Circuit findings in a resident–intruder assay identify mechanisms of a particular animal interaction rather than predicting the behavior of an individual person. Explaining the tested action precisely preserves the distinction between a mechanistic experiment and a broad label applied across very different situations.



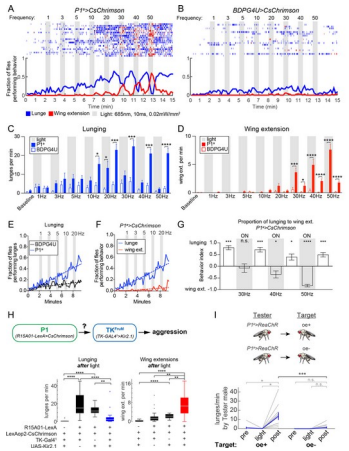
Carlson and Birkett (2022), Fig. 11.7, p. 345.

A fly circuit can create a lasting bias toward aggression

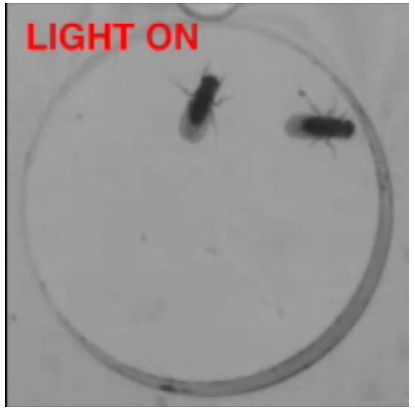
A fruit-fly study manipulated **P1 interneurons**, a small neural population also associated with male courtship. The researchers tested whether activation could enhance aggression after stimulation ended. This design distinguished a persistent internal bias from a movement that occurs only while a light-sensitive channel or temperature-sensitive actuator is active.

Activating selected P1 neurons increased later fighting under the tested conditions. The effect could persist for minutes and influence how males responded when a barrier was removed. The circuit manipulation therefore altered subsequent social responsiveness rather than only causing an immediate lunge independent of another fly.

The same population's participation in courtship and aggression did not make those behaviors identical. Different downstream conditions and levels of activation could favor different actions. Studying the sequence of responses helped identify a shared internal drive whose overt expression depended on the current encounter and the strength of neural activation.



Hoopfer et al. (2015), Fig. 3.



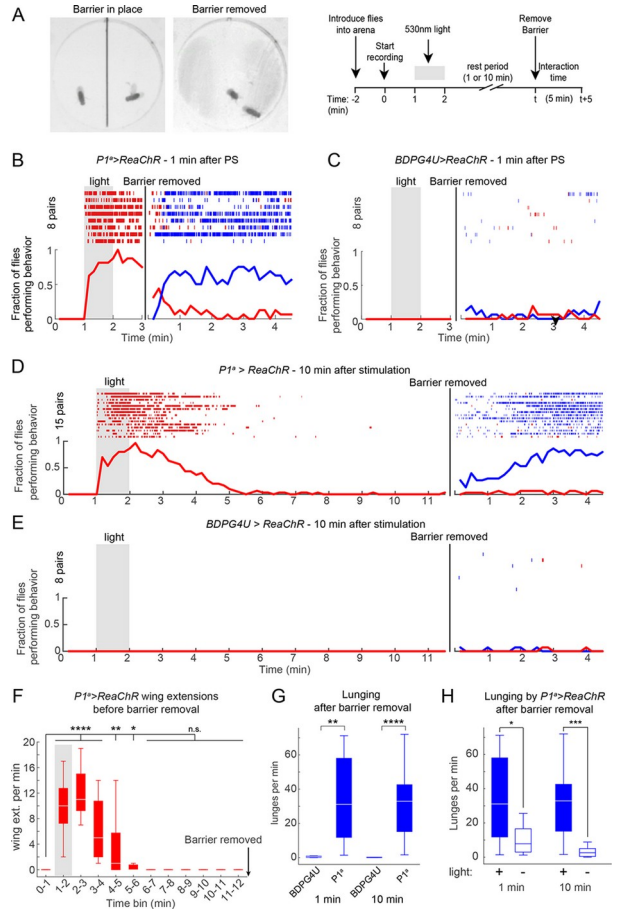
Hoopfer et al. (2015), media4.

Activation strength can favor different social actions

The P1 study varied optical stimulation and compared aggression with unilateral wing extension, a component of fly courtship. At lower stimulation levels, aggression could be promoted without strong wing extension. Stronger drive favored wing extension during stimulation and changed the expression of fighting across the stimulation period.

After strong stimulation ended, aggression could re-emerge while wing extension declined. A persistent state and the action expressed at that moment therefore followed different time courses. The absence of fighting during the strongest stimulation did not mean the manipulation had removed every internal condition that later facilitated aggression.

The investigators also tested aggression when normal wing display was unavailable. These comparisons separated one motor act from the state influencing social behavior. The resulting circuit account explains how a population can bias several actions without each action being a direct, obligatory copy of the same stimulation waveform.



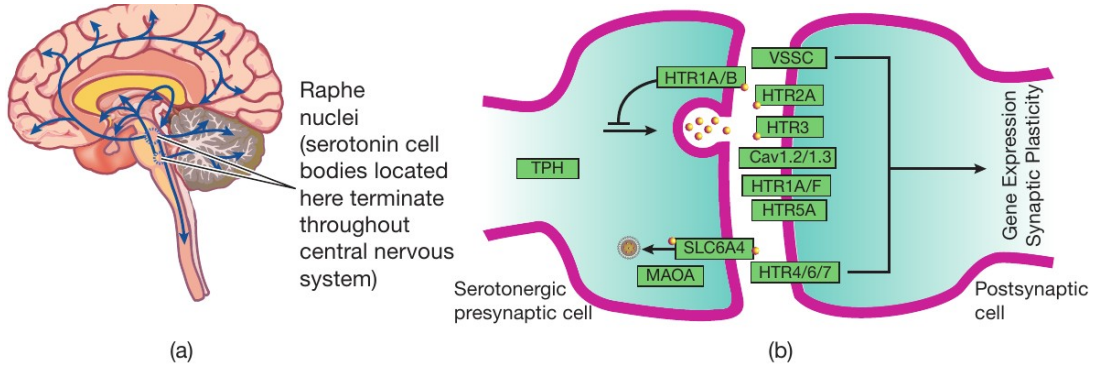
Hoopfer et al. (2015), Fig. 5.

Serotonin acts through several receiving receptors

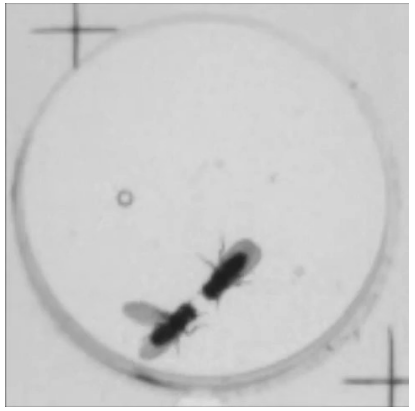
Serotonin is a transmitter released by neurons concentrated in brainstem raphe nuclei and distributed to many targets. It acts through several receptor subtypes with different cellular effects. A change in serotonin signaling can therefore alter excitability, inhibition, or synaptic regulation differently across the circuits controlling a social encounter.

Animal studies associate particular serotonergic manipulations with changes in impulsive or aggressive actions, but the direction depends on receptor, site, timing, and task. Measuring a serotonin breakdown product in a bodily fluid is also different from measuring transmitter release at a specific brain synapse during an attack.

Impulsivity concerns acting without sufficient adjustment to consequences or stopping demands; it is not synonymous with every form of aggression. Planned aggression and rapid reactive attack can depend on different processes. A mechanistic account identifies the receptor-controlled circuit and the specific behavioral component altered rather than labeling serotonin as an aggression switch.



Carlson and Birkett (2022), Fig. 11.9, p. 347.



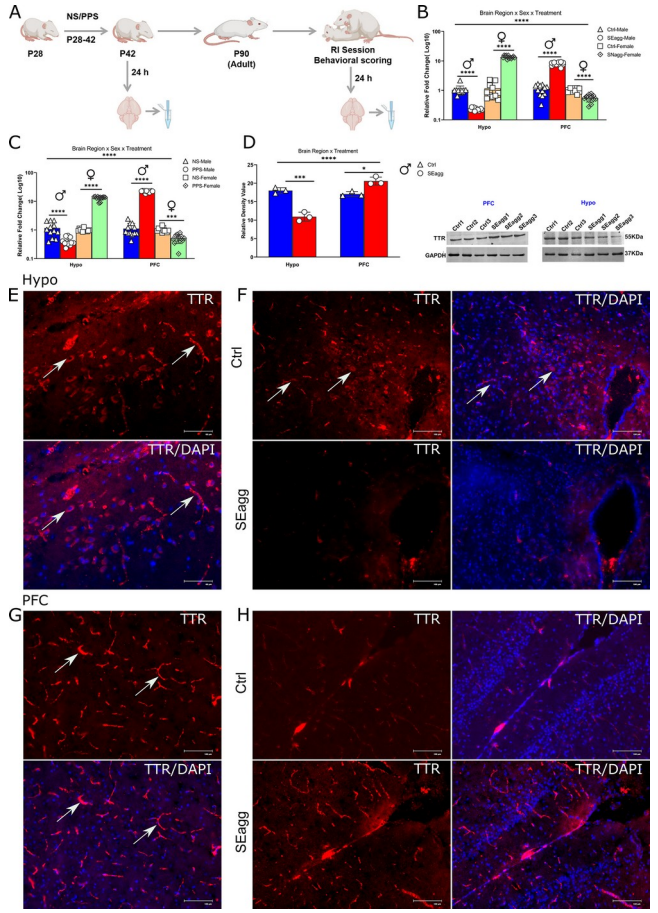
Hoopfer et al. (2015), media1.

Early stress changes local biology and later aggression

A mouse study examined consequences of stress during the period around puberty. Adult males exposed to the stress procedure showed escalated aggressive behavior and region-specific changes in **transthyretin**, or TTR, a protein involved in carrying thyroid hormone. The study connected a developmental experience to a later molecular and behavioral phenotype.

TTR expression changed differently in hypothalamus and prefrontal cortex, while circulating thyroid hormone did not show an equivalent global change. Local brain hormone availability could therefore differ from the signal suggested by a blood measurement. A normal circulating level did not establish that every neural target was experiencing the same thyroid-related conditions.

The researchers also measured chemical marks regulating gene expression in the affected regions. These observations described a lasting response to experience rather than a change in the DNA sequence inherited by every cell. The specific mouse stress paradigm should not be treated as a universal explanation for all aggression after human adversity.



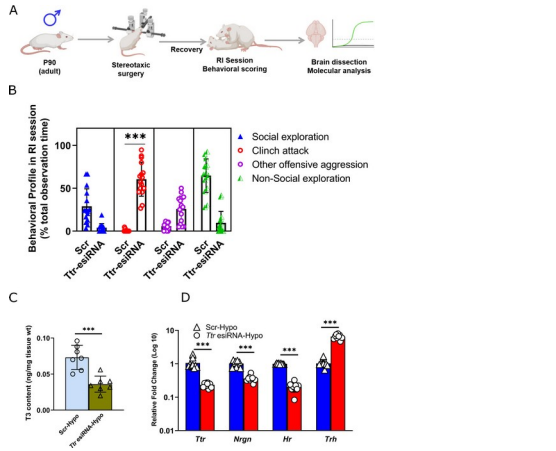
Rawat et al. (2022), Fig. 3.

Reducing a local transporter can increase aggression

The early-stress study tested whether lowering hypothalamic TTR could reproduce aspects of the behavioral effect. **RNA interference** reduces production of a target protein by interfering with its RNA. This manipulation changed local expression without requiring the animal to experience the earlier stress procedure itself.

TTR reduction increased aggression in the tested male mice and reduced local thyroid-hormone availability and responses of thyroid-sensitive genes. The experiment therefore moved beyond a correlation between stress history and transporter expression. A defined molecular intervention altered both the local biological environment and a measured pattern of social action.

The findings differed across sexes and brain regions, emphasizing the conditions under which the effect was established. TTR also has functions outside this behavioral pathway. The demonstrated mechanism concerns local hormone handling within an experimental circuit, rather than proving that one transporter measurement can diagnose or predict violent behavior in a person.



Rawat et al. (2022), Fig. 5.



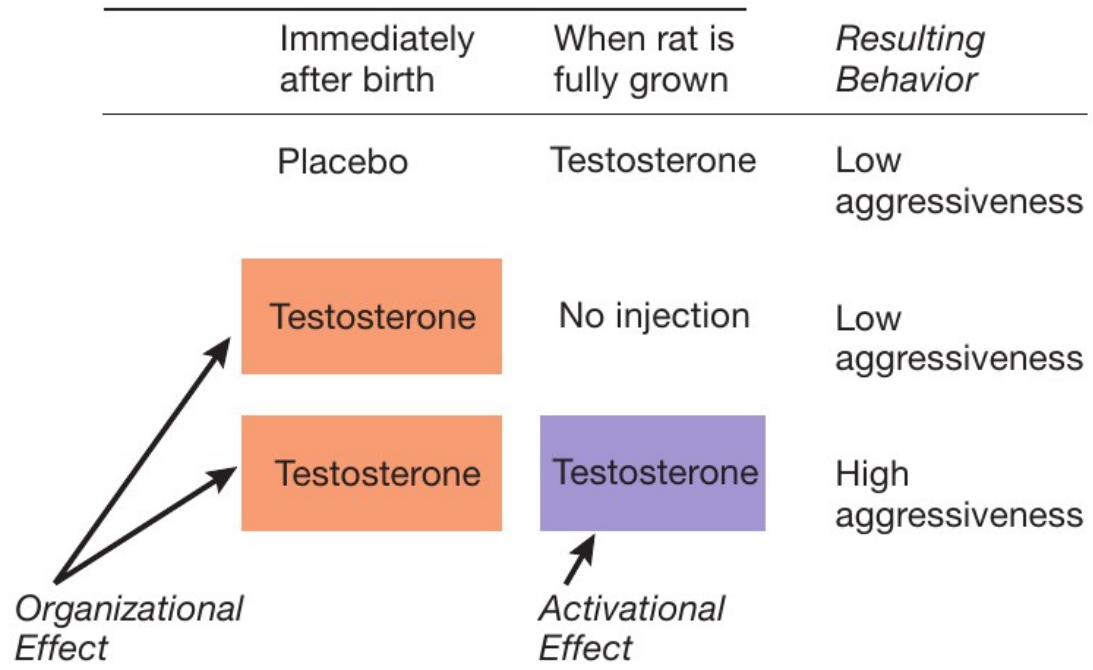
Rawat et al. (2022), Video 2.

Hormones shape developing and adult aggression

Androgens and estrogens act in neural populations involved in social behavior, including aggression. **Organizational effects** alter development or lasting responsiveness, while **activational effects** change the operation of an existing system. The behavioral outcome can depend on both the animal's developmental history and its endocrine condition when the encounter occurs.

In rodent experiments, removing gonadal hormones and replacing particular steroids can change aggressive responses under defined social conditions. These interventions affect sensory processing, motivation, and receptor-dependent signaling as well as the final action. A reduction in attack after gonad removal does not establish that testosterone directly commands each aggressive movement.

Cell populations responsive to steroids can participate in mating, territorial behavior, defense, or caregiving according to their connections and current state. Hormone action therefore needs an identified neural target and a measured endpoint. The same circulating compound can have different consequences in two cells with different receptors or projection targets.



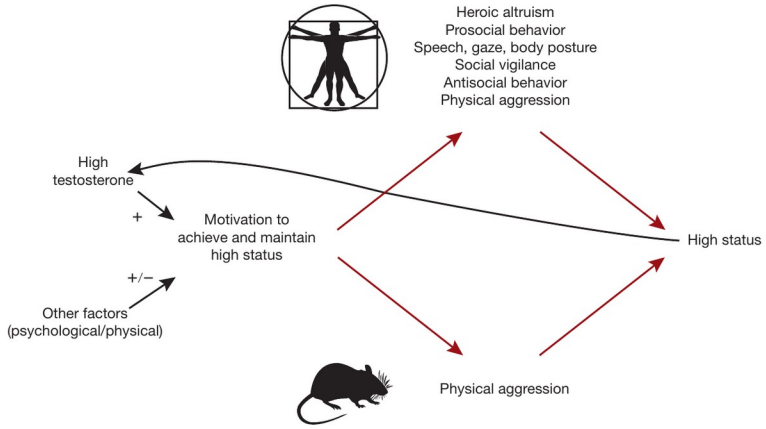
Carlson and Birkett (2022), Fig. 11.13, p. 349.

Testosterone effects depend on social context

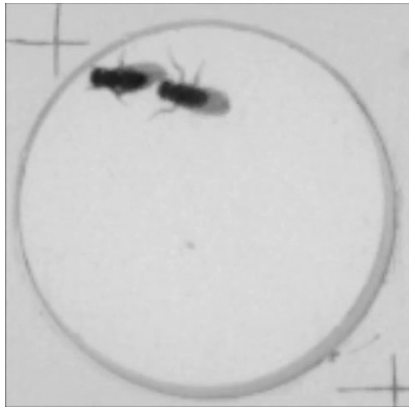
Testosterone is associated with responses to competition and status in several species, but high status can be obtained through different actions. Human cooperation, generosity, skill, and intimidation can each affect social standing under different conditions. A hormone associated with status-related motivation therefore need not uniformly increase physical aggression.

Hormone concentration can also change after a competitive outcome, making the direction of causation important. A pre-existing endocrine difference and a response to winning or losing are different hypotheses. Correlations between testosterone and behavior cannot separate them without a design that measures timing or experimentally changes the relevant signal.

Context dependence means that the consequence of a signal changes with the available actions and their expected outcomes. The same endocrine state can bias different strategies in different environments. Interpreting testosterone requires identifying the social situation, the behavioral measure, and whether the evidence concerns association or a controlled administration effect.



Carlson and Birkett (2022), Fig. 11.14, p. 351.



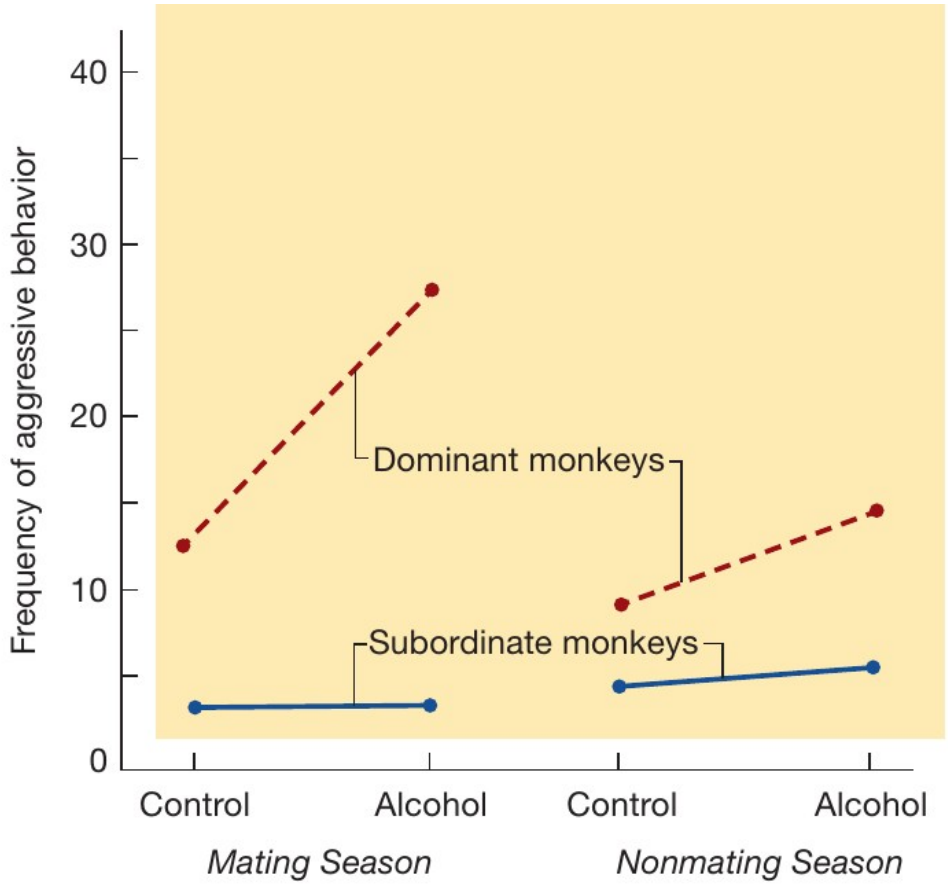
Hoopfer et al. (2015), media2.

Alcohol can alter existing social tendencies

Alcohol affects several neural signaling systems and can change attention, judgment, and behavioral inhibition. Its effect on aggression depends on dose, prior behavior, social context, and the response being measured. An intoxicated animal or person does not express an identical aggressive response in every setting simply because alcohol has entered the bloodstream.

In animal studies, dominance relationships and reproductive conditions can influence the effects of alcohol on aggressive encounters. A treatment may amplify a behavior in one social group while producing a smaller change in another. This interaction differs from a universal rule that the substance creates the same aggressive motivation in every individual.

Human observations also combine pharmacological effects with expectations, provocations, opportunities, and learned behavior. These contributors can be studied separately with controlled designs. A circuit explanation must identify which process changed, such as response inhibition or threat interpretation, rather than treating the drug name as a complete explanation of the resulting action.



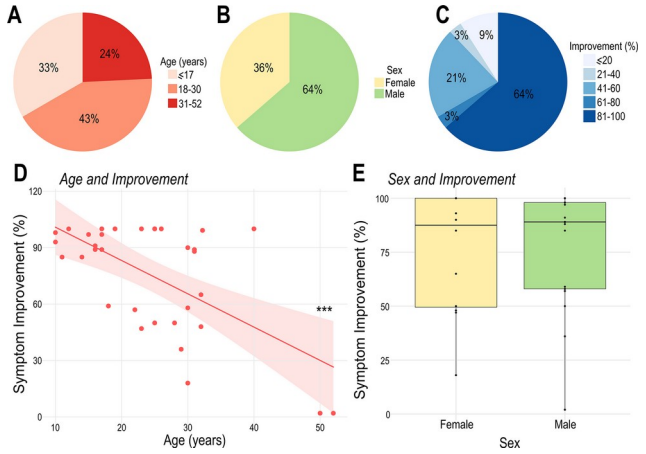
Carlson and Birkett (2022), Fig. 11.15, p. 352.

Stimulation studies examine severe clinical aggression

A multicenter study analyzed hypothalamic deep-brain stimulation in people treated for severe, difficult-to-manage aggressive behavior. **Deep-brain stimulation** applies electrical pulses through implanted electrodes. The study examined differences in stimulation location and the networks associated with clinical change, rather than assuming that stimulating the entire hypothalamus has one uniform effect.

Treatment outcomes varied, and electrode-related maps helped identify locations associated with greater improvement in the studied cases. Such analyses concern a selected clinical population receiving an invasive intervention. They do not establish a general biological treatment for ordinary anger or a method for predicting aggression in unselected people.

Clinical stimulation can influence axons as well as nearby cell bodies, so the affected network extends beyond a point on a scan. The anatomical analysis therefore asked which connected systems were recruited. This provides a more specific interpretation than describing a single aggression center that simply becomes inactive when an electrode is turned on.



Gouveia et al. (2023), Fig. 2.



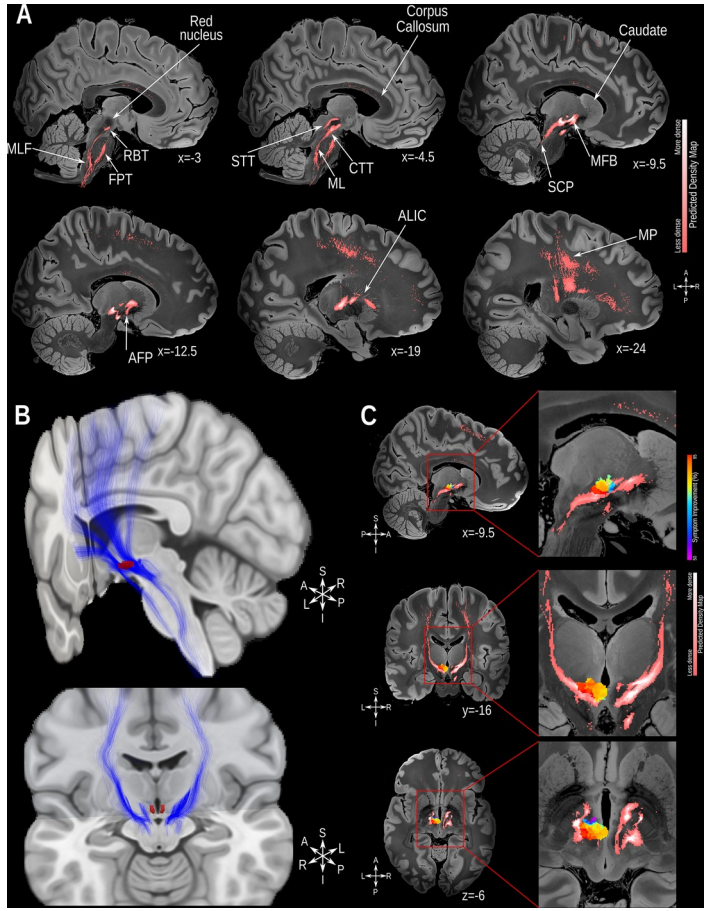
Rawat et al. (2022), Video 1.

Clinical effects depend on the network being stimulated

The hypothalamic stimulation study combined electrode localization with structural and functional connectivity analyses. **Structural connectivity** estimates anatomical pathways, while **functional connectivity** describes statistical coordination of activity. These measures ask different questions and do not both demonstrate direct synaptic communication between every pair of regions they identify.

Patterns associated with clinical improvement involved distributed networks rather than the electrode's immediate neighborhood alone. The researchers also compared spatial network patterns with gene-expression resources to generate molecular hypotheses. These associations can guide further investigation without showing that a particular gene was directly changed in every treated patient.

The study's causal intervention was electrical stimulation, while many explanatory maps were correlational or model-based. Keeping those evidence types separate preserves what was demonstrated: clinical outcomes varied with the network affected. It also prevents turning a population-derived connectivity pattern into a universal explanation for every aggressive act.



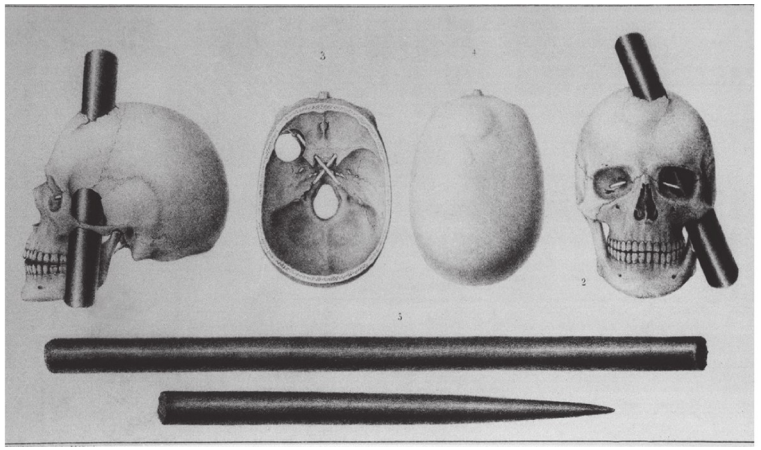
Gouveia et al. (2023), Fig. 4.

Prefrontal circuits help adjust actions to their consequences

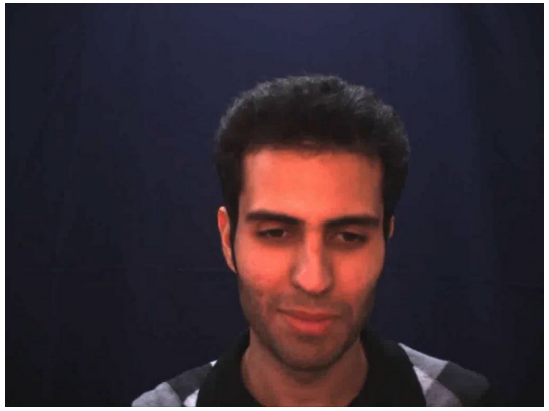
The **ventromedial prefrontal cortex**, or vmPFC, interacts with amygdalar, striatal, and other cortical systems involved in value and action. Damage can impair adjustment to social consequences even when a person retains language and many forms of reasoning. Emotional regulation therefore involves using outcome information to guide behavior, not merely suppressing every emotional response.

Impulse control requires maintaining goals, evaluating consequences, and changing an action when circumstances demand it. These operations depend on distributed networks rather than one inhibitory center. A deficit can appear in learning from outcomes, representing future costs, or stopping a response, with different consequences for the observed behavior.

Historical lesion cases helped identify these functions but often involved extensive damage and incomplete measurements. Modern experiments can isolate more specific circuit and task variables. The useful question is which computation or action changed after injury, rather than whether a person's entire personality or moral character occupies a single cortical location.



Carlson and Birkett (2022), Fig. 11.16, p. 354.



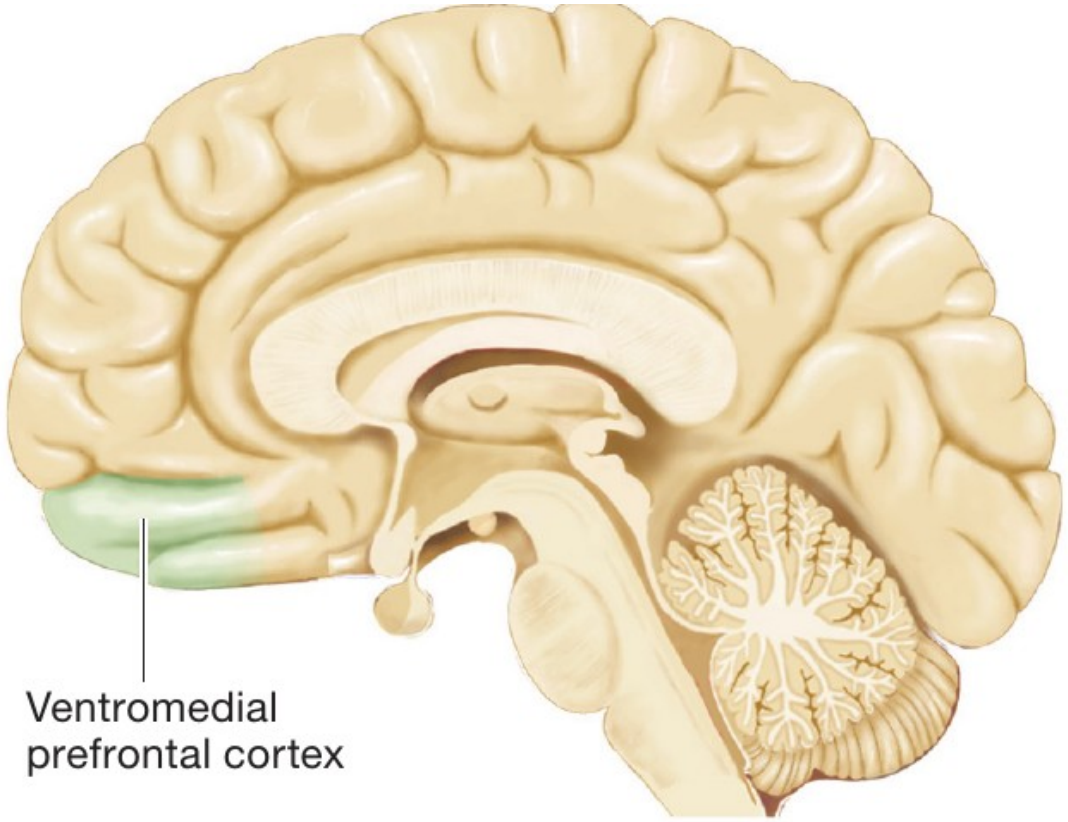
Jeganathan et al. (2022), Fig. 3-Video 1.

Control networks continue changing across development

Prefrontal and subcortical networks develop through changes in connectivity, synaptic organization, and experience-dependent learning. Different abilities mature along different trajectories, and individuals vary. Brain development is therefore not a switch that changes every person from emotionally uncontrolled to fully mature on one universally fixed birthday.

Developmental studies compare task performance and neural measurements across ages while considering factors such as motivation and social context. A young participant may regulate behavior effectively in one situation but respond differently when peers, rewards, or uncertainty change. Age alone does not specify the complete state of the control system during each decision.

Learning also continues in adulthood, and mature networks remain sensitive to stress, sleep, drugs, and experience. Developmental differences identify changing capacities and conditions rather than an absence of responsibility or a guarantee of perfect control. The biological account concerns interacting processes whose timing must be measured rather than assumed from a simple age label.



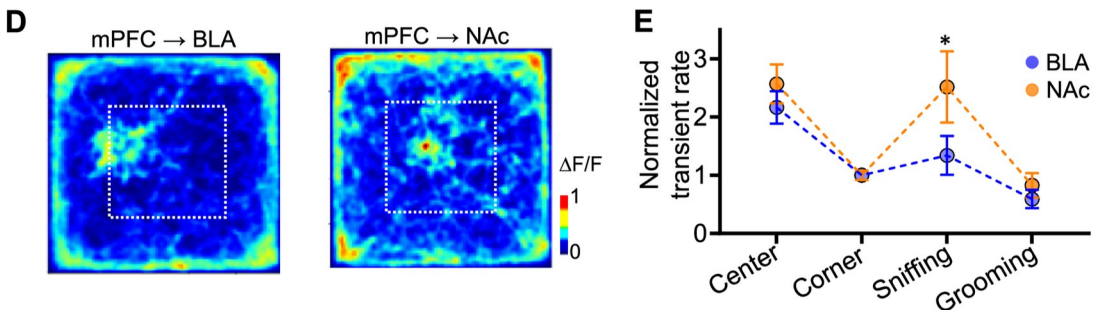
Carlson and Birkett (2022), Fig. 11.6, p. 342.

Prefrontal outputs represent different affect-related states

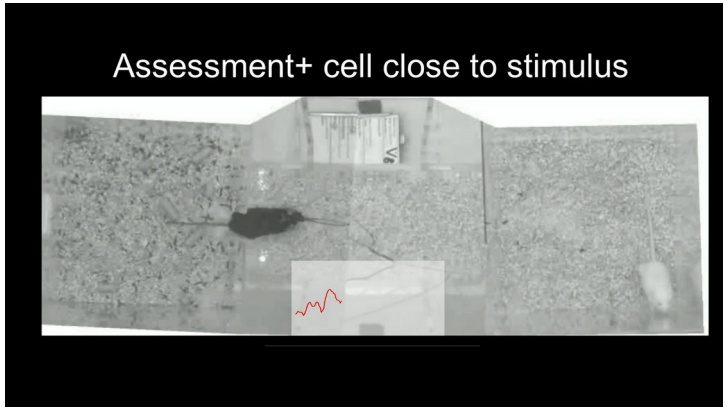
A recent mouse study compared medial-prefrontal neurons projecting to basolateral amygdala with those projecting to nucleus accumbens. Projection labels identified the destination of a neuron's axon rather than assuming every cell in the same cortical region had the same role. The researchers recorded population activity during several behaviorally defined situations.

Amygdala-projecting cells showed patterns associated more strongly with anxiety-related conditions, while accumbens-projecting cells were more active during exploratory and approach-related behavior. These differences became clearer when specific neuronal subsets were examined. Averaging all prefrontal activity together could conceal functional variation among its output pathways.

The investigators identified **center-ON neurons**, cells whose activity increased around entry into an exposed central area. These cells were more prevalent in the amygdala-projecting population and differed from cells responding during grooming or sniffing. Selecting a behaviorally defined subset therefore exposed information that was diluted when every recorded neuron was averaged together.



Lai et al. (2026), Fig. 2.



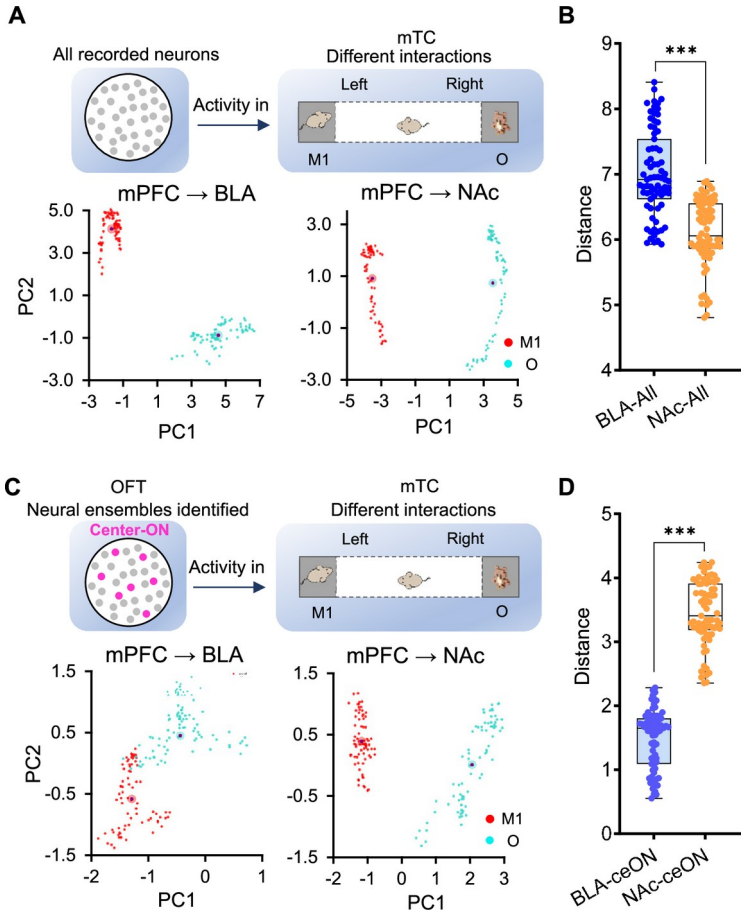
Krzywkowski et al. (2020), Video 1.

Population patterns change beyond average activity

The prefrontal-projection study compared how distributed activity patterns changed across contexts. **Pattern decorrelation** means that the relative activity across neurons becomes less similar between conditions. Two populations can have comparable average activity while their individual cells rearrange which information is most strongly represented.

Among center-ON neurons, the accumbens-projecting subset separated social from object interactions more strongly. Across all recorded neurons, however, the amygdala-projecting population showed greater separation. The result depended on which cells were analyzed: a selected subset and an entire output population could organize the same comparison differently rather than producing one interchangeable regional signal.

The investigators also examined behavioral consequences of repeated winning and losing in a separate social-ranking task. These experiences changed measured state variables and stress-hormone levels. The comparison provided biological context for the defined conditions without implying that every social outcome is represented identically across both prefrontal output routes.



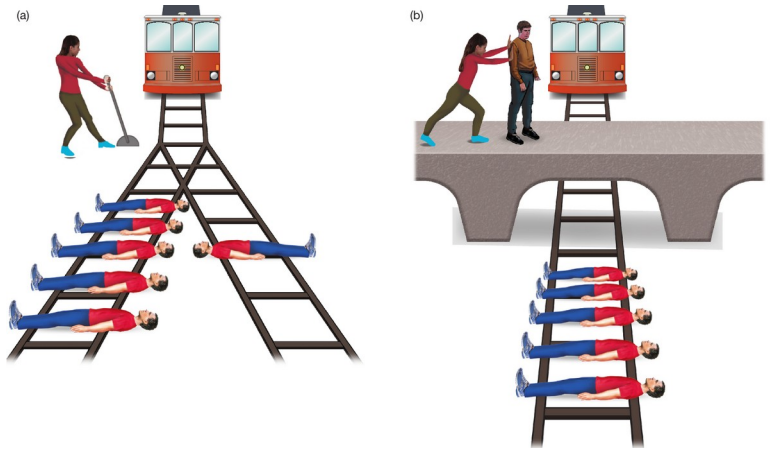
Lai et al. (2026), Fig. 5.

Moral decisions combine value, responsibility, and action

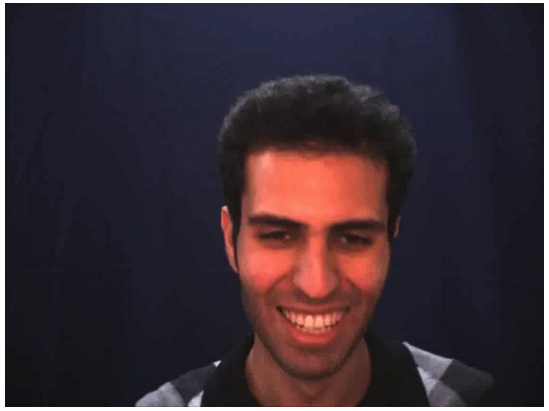
Moral-choice experiments ask how people weigh consequences, intentions, obligations, and personal involvement. **A moral dilemma** places competing considerations in conflict, making their relative influence measurable. Responses depend on wording, framing, and the participant's interpretation, so one hypothetical choice is not a complete measure of everyday moral behavior.

Lesion and imaging studies implicate networks involved in valuation, emotional response, social knowledge, and cognitive control. These processes can contribute differently to an impersonal judgment and an action involving direct harm. The difference does not mean that one decision is purely emotional while the other is produced without any affective input.

A neural measurement can identify processing associated with a particular task comparison without classifying an individual as good or bad. Moral reasoning and real-world conduct require broader evidence. The mechanistic question concerns how information about outcomes and other people changes a decision, not where a universal morality center is located.



Carlson and Birkett (2022), Fig. 11.19, p. 357.



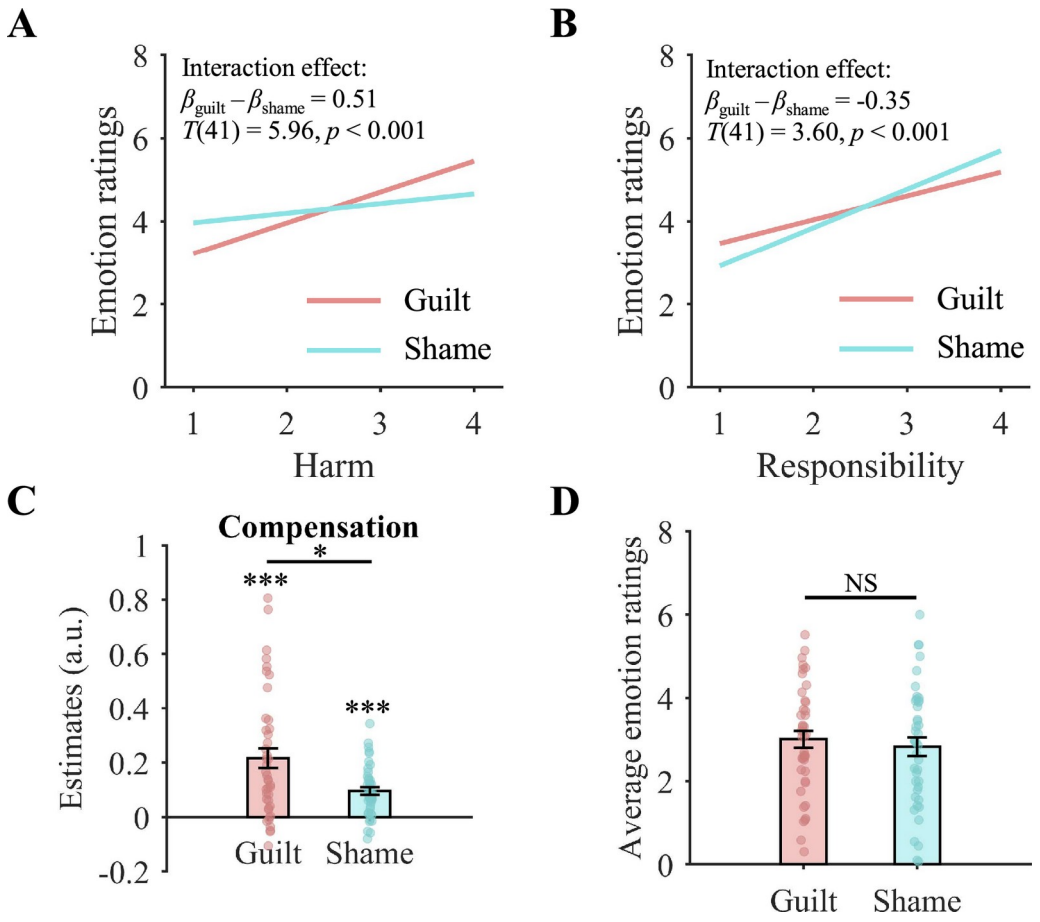
Jeganathan et al. (2022), Fig. 3-Video 2.

Harm and responsibility shape guilt and shame differently

A human experiment independently manipulated harm to another person and responsibility for that outcome. Participants reported **guilt** and **shame** and could make compensatory choices. Separating these antecedents allowed the investigators to ask which information influenced each reported emotion rather than treating the two words as interchangeable labels for general unpleasantness.

Harm had a stronger effect on guilt in the task, while responsibility had a stronger effect on shame. Guilt also exerted a stronger influence on compensation under the tested conditions. The distinction concerned measured sensitivities within the experiment, not a claim that every person always experiences the two emotions in exactly the same way.

The researchers measured both feeling reports and decisions about helping the affected person. These outcomes can vary separately: a report of an unpleasant emotion does not automatically determine the amount of compensation. Linking the two required examining how changes in harm and responsibility influenced the subsequent behavioral choice.



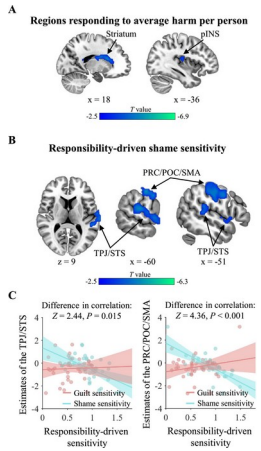
Zhu et al. (2025), Fig. 2.

Models separate emotional sensitivity from compensation

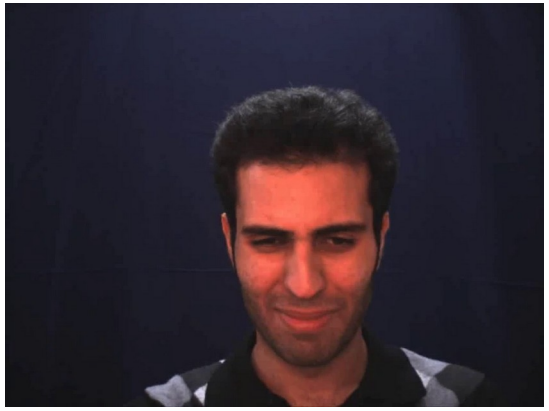
The guilt-and-shame study used **computational modeling** to describe how participants integrated harm and responsibility. A model makes assumptions explicit in equations that can be compared with observed choices. It does not directly reveal a person's private reasoning unless its predictions are supported by the measured behavior.

Imaging implicated regions involved in value and social information, with differences associated with guilt sensitivity, shame sensitivity, and compensation. For example, responsibility-related shame variation involved mental-state-related regions, while shame-related compensation involved lateral prefrontal activity. Different stages of the task therefore did not reduce to one identical activation map.

The model also captured patterns consistent with diffusion of responsibility when responsibility was shared. This supplied a specific description of how antecedent information affected emotion and action. The neural results were associations with modeled variables rather than proof that a single activated region alone generated guilt, shame, or a compensatory decision.



Zhu et al. (2025), Fig. 4.



Jeganathan et al. (2022), Fig. 3-Video 3.

Emotional expressions have biological and cultural inputs

Facial muscles produce recognizable movements through neural motor pathways. Similarities across development and cultures support biological contributions to expression, but how expressions are used and interpreted also depends on experience. A posed face selected from a small set of labels is therefore a different stimulus from an unfolding social interaction.

Display rules are learned patterns affecting when and how people express emotion in a particular social setting. Someone can suppress a smile, exaggerate surprise, or use a polite expression while feeling something different. An expression is consequently a communicative action rather than an infallible window into one private emotional state.

Cross-cultural experiments must consider the response options, verbal instructions, familiarity of the portrayed situation, and whether the expression was spontaneous or deliberately posed. Shared recognition of some facial actions does not imply perfect universality for every emotion word. The relevant biological claim concerns particular movements and conditions, not an identical emotional vocabulary across all societies.



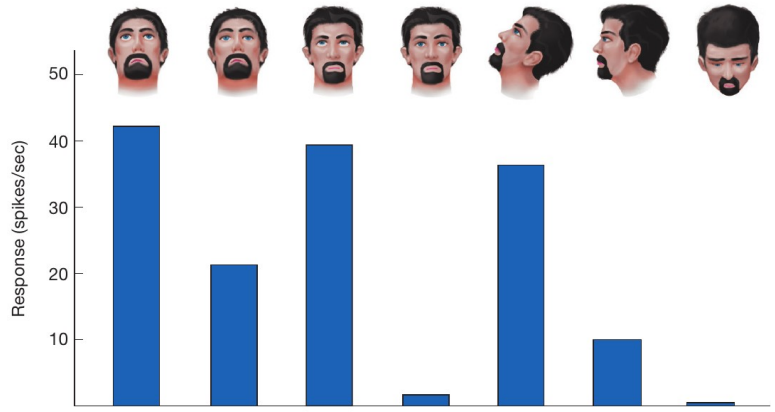
Carlson and Birkett (2022), Fig. 11.20, p. 359.

Gaze, posture, and context change facial interpretation

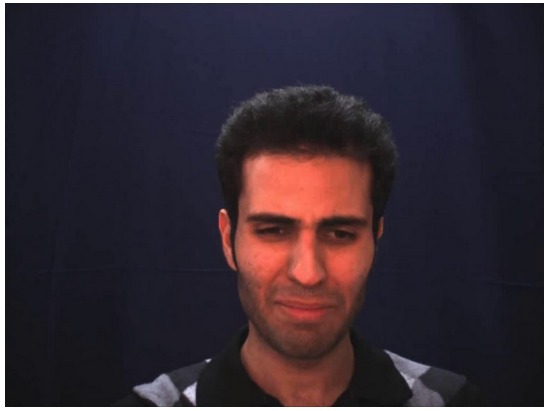
A facial movement changes over time and occurs within a social situation. Gaze direction can indicate the target of attention, while body posture can suggest approach or withdrawal. **Context** changes what an observer infers from the same face, including whether a smile signals enjoyment, embarrassment, politeness, or an attempt to reassure.

Neurons in temporal social-perception networks respond to information such as face configuration, motion, and gaze direction. Those signals must be integrated with knowledge about the scene and the other person's actions. Recognizing where someone is looking is therefore different from determining exactly what they feel or intend.

An observer can also use movement timing rather than only the final shape of an expression. A brief change around the eyes or mouth can alter the interpretation of a longer interaction. Experiments using dynamic stimuli examine information that is lost when a face is reduced to one carefully selected still photograph.



Carlson and Birkett (2022), Fig. 11.27, p. 364.



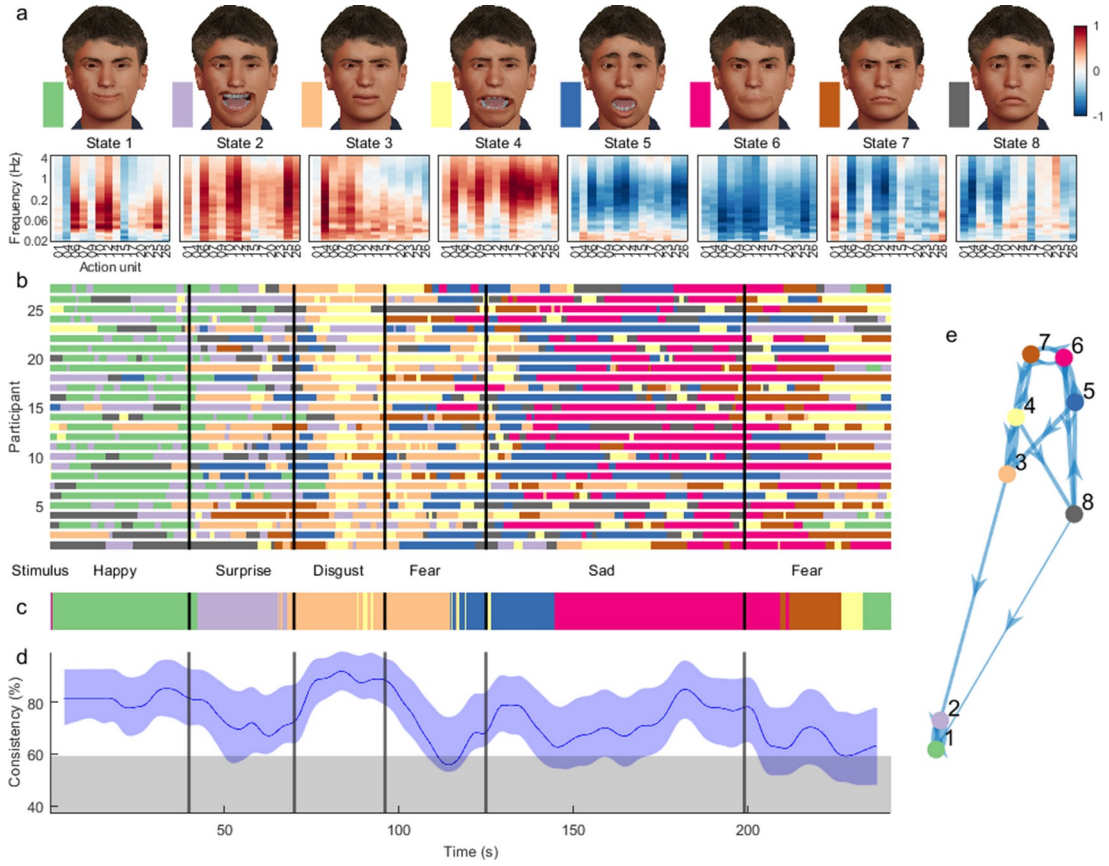
Jeganathan et al. (2022), Fig. 3-Video 4.

Dynamic facial actions can be described as changing states

A human study recorded faces while people watched emotionally evocative films. The researchers measured **facial action units**, recognizable movements such as pulling the lip corners or tightening part of the face. Combining these actions through time preserved changes in movement rather than assigning each video frame one predetermined emotion word.

Analysis identified recurring spatiotemporal states composed of multiple facial actions with characteristic dynamics. People viewing the same film could pass through related sequences, even though their expressions were not identical at every moment. The study therefore modeled shared organization in the timing and combination of movements rather than only the intensity of a smile.

The accompanying source recordings show actual participants' facial behavior grouped by an analysis-defined state. A state is a mathematical description of observed motion, not direct proof of one universal subjective emotion. The distinction allows the measured expression to remain useful without claiming the camera can read every private feeling.



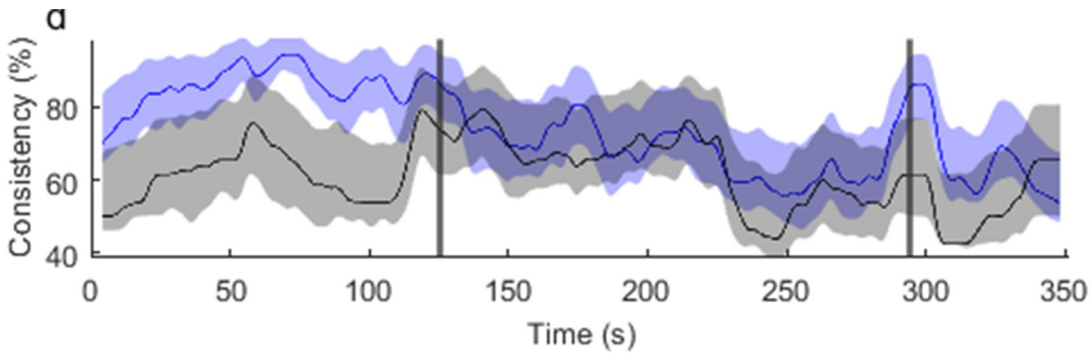
Jeganathan et al. (2022), Fig. 3.

Depression can alter facial dynamics during a film

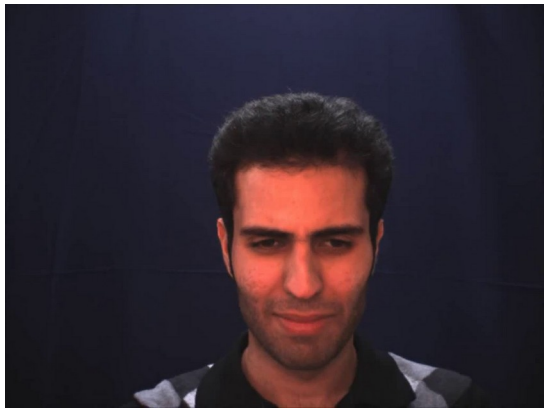
The dynamic-expression study compared controls with participants who had the melancholic subtype of major depressive disorder. **Melancholia** refers to a defined clinical pattern within depression, not simply a momentary sad mood. Participants were studied with the same kinds of film-evoked facial measurements so that the temporal patterns could be compared.

The groups differed in aspects of facial-action dynamics and inferred state sequences. The analysis detected information that a single static photograph or overall average movement could miss. The result concerned patterns across recorded participants rather than a diagnostic rule that every person with depression must show one particular expression.

Clinical variation can affect movement frequency, coordination, and responsiveness, while medication and other factors may also influence measured behavior. The study introduced a quantitative description of these dynamics. It did not establish that an analysis state itself was the cause of depression or that observation of one short facial clip could replace a clinical assessment.



Jeganathan et al. (2022), Fig. 6.



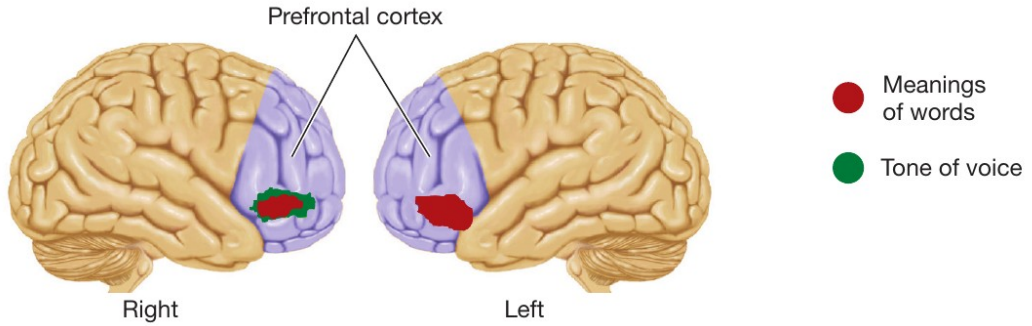
Jeganathan et al. (2022), Fig. 3-Video 5.

Tone of voice and word meaning use distinct processes

Emotional prosody is the pattern of pitch, timing, and intensity through which a voice conveys affective information. It can change the meaning of the same words, such as turning a statement into reassurance or sarcasm. Understanding the words and interpreting their tone therefore require related but distinguishable forms of processing.

Lesion and imaging evidence often shows greater right-hemisphere involvement in aspects of emotional prosody, while many language operations are more left-lateralized. These are relative tendencies within distributed networks, not a complete division in which one hemisphere feels and the other thinks. Both sides participate in several components of communication.

Aprosodia is impaired production or understanding of prosodic information. A person can retain the ability to understand sentence meaning while struggling with its emotional tone. Comparing these functions after damage helps distinguish acoustic, linguistic, and social interpretation rather than treating every communication difficulty as a loss of all emotional understanding.



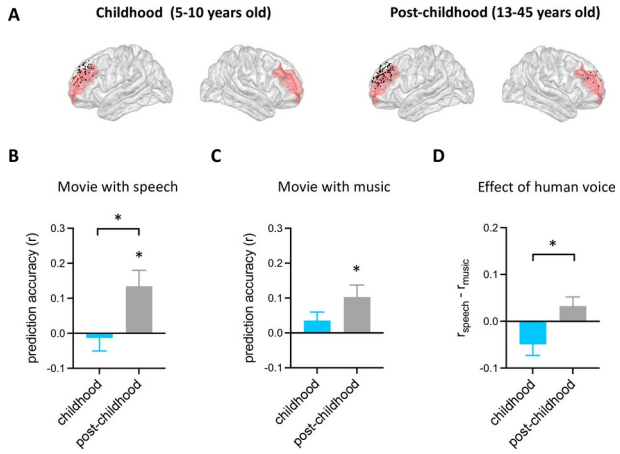
Carlson and Birkett (2022), Fig. 11.21, p. 360.

Facial-emotion representations change across development

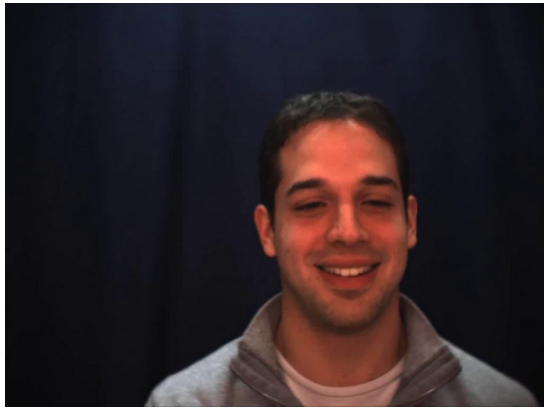
A recent human study used **intracranial EEG**, recordings from electrodes placed inside the skull for clinical reasons, to study facial-expression processing across age groups. This method provides precise timing at sampled locations, but coverage depends on clinical placement rather than a complete, uniform survey of every brain region in every participant.

Posterior superior temporal cortex represented facial-expression information in both younger and older groups. Dorsolateral prefrontal representations were more evident in the post-childhood group under the tested analysis. Development therefore changed the distribution of measurable information across the sampled network rather than simply switching facial perception on at one age.

The investigators related neural responses to features of facial emotion and participant age. The result concerned the representation of information, not direct control of the facial muscles. It helps distinguish recognizing another person's expression from generating an expression oneself, two processes that use interacting but nonidentical pathways.



Fan et al. (2026), Fig. 2.



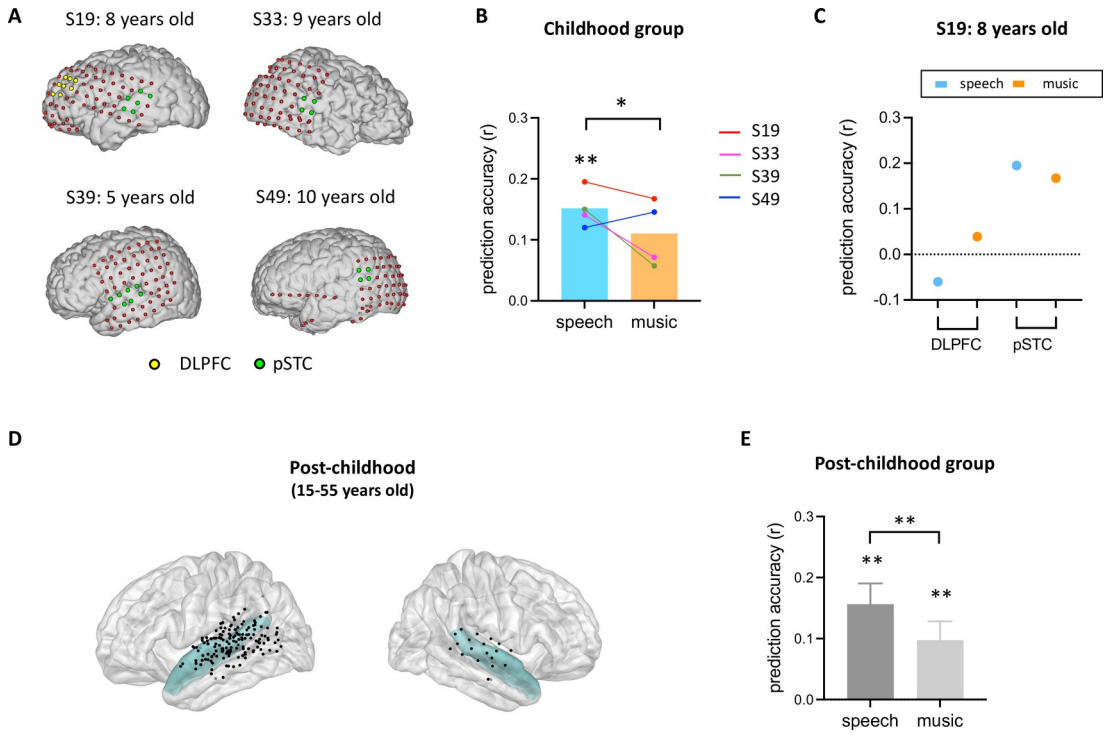
Jeganathan et al. (2022), Fig. 3-Video 7.

Expression information expands beyond sensory regions

The developmental intracranial study used encoding models to test which aspects of facial expression predicted neural responses. **An encoding model** describes how measured stimulus features relate to activity. Successful prediction indicates that the sampled population contains information related to those features, rather than proving it independently understands every social meaning of the expression.

With increasing age, representation of complex emotional features became more pronounced in posterior temporal cortex, while older participants also showed prefrontal encoding. The pattern was consistent with development of a broader processing network. Neural responses differed by region and feature rather than following one identical maturation curve across the entire cortex.

The film alternated blocks with dialogue and blocks accompanied by music. In older participants, human voice enhanced prefrontal representation of facial-expression features, while younger participants showed a different prefrontal effect. Posterior temporal responses also depended on the accompanying sound, indicating that audiovisual context influenced how expression information was represented across the sampled network.



Fan et al. (2026), Fig. 3.

The amygdala can influence where an observer looks

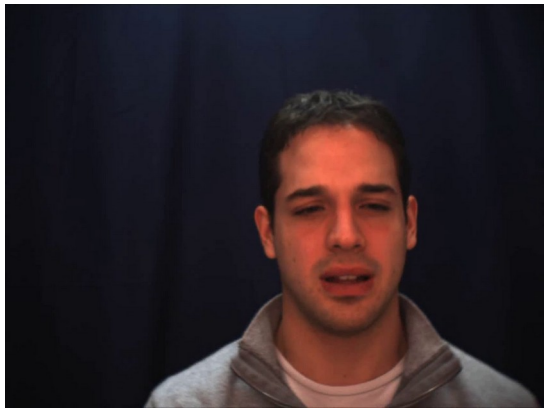
People with amygdala damage can have difficulty recognizing fearful faces partly because they sample different facial information. **Eye fixation** is a period during which gaze is directed toward a particular location. Recording fixations reveals what information was visually available before deciding whether the observer could interpret the expression.

In a well-studied lesion case, attention to the eye region was reduced during spontaneous viewing. Directing attention toward the eyes improved recognition of fearful expressions. The deficit therefore involved how diagnostic information was acquired, rather than proving that every representation needed to identify fear had been completely destroyed.

The eye region contains cues such as widened lids and visible sclera, the white portion of the eye. Those features matter in combination with the rest of the face and context. Amygdala-related processing can therefore contribute to selecting useful information as well as to reactions after a stimulus has already been recognized.



Carlson and Birkett (2022), Fig. 11.26, p. 363.



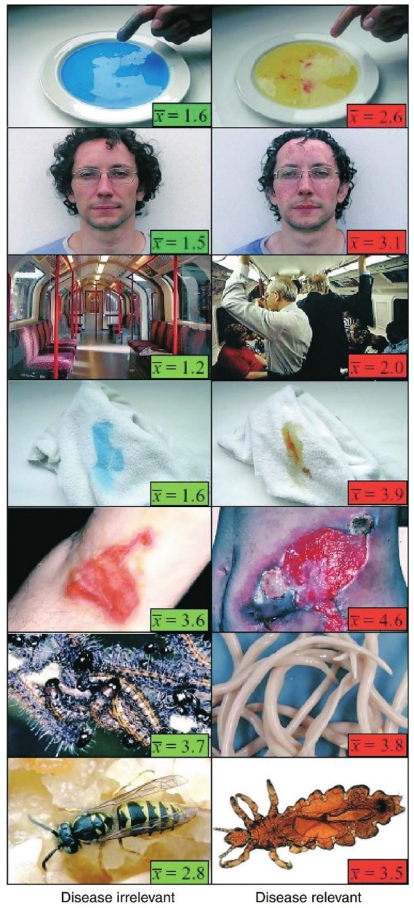
Jeganathan et al. (2022), Fig. 3-Video 8.

Disgust involves bodily significance and recognition

Disgust often promotes avoidance of contamination, spoiled food, or other potentially harmful contact. It can involve nausea, changes in facial action, and subjective revulsion. These components are related but distinct, and similar imagery can produce different reactions depending on experience, meaning, and the situation in which it is encountered.

The **insula** participates in processing bodily signals and the affective significance of events, including many disgust-related tasks. Its activity is not unique to disgust: the region also contributes to other sensory, motivational, and social processes. Observing an insular blood-oxygen response therefore does not identify a person's exact emotion without additional evidence.

Experiments can compare contamination-related images with visually similar but noncontaminating scenes to separate physical image properties from inferred biological significance. The response can depend on what the observer believes the material is. Perception of a disgusting object consequently involves both the available sensory features and knowledge about their potential consequences for the body.



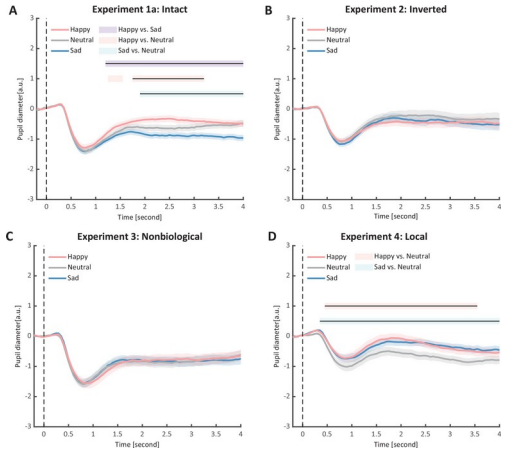
Carlson and Birkett (2022), Fig. 11.29, p. 366.

Body motion can produce emotion-related pupil responses

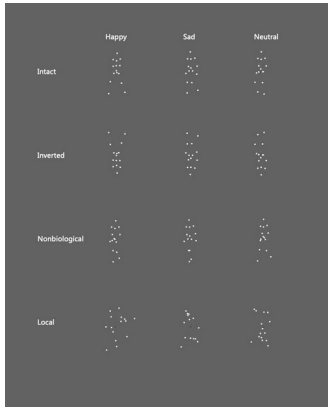
A human study presented **point-light walkers**, moving dots that preserve aspects of body-joint motion while removing most appearance information. Participants viewed movements conveying happiness, sadness, or a neutral condition. The experiment tested whether motion alone influenced physiological responses rather than requiring a detailed facial expression or realistic clothing.

Pupil size differed across the motion conditions: intact happy movement produced greater dilation than the neutral and sad comparisons under the tested conditions. **Pupil dilation** is an autonomic response influenced by several processes, including attention and arousal. Its difference across conditions did not uniquely identify every aspect of the observer's subjective feeling.

The stimulus retained the organization of a moving body while controlling much of its visual appearance. It therefore isolated information often overlooked in face-only emotion tasks. The result showed that social movement could influence bodily response even when the display provided no visible face from which to read an expression.



Yuan et al. (2024), Fig. 2.



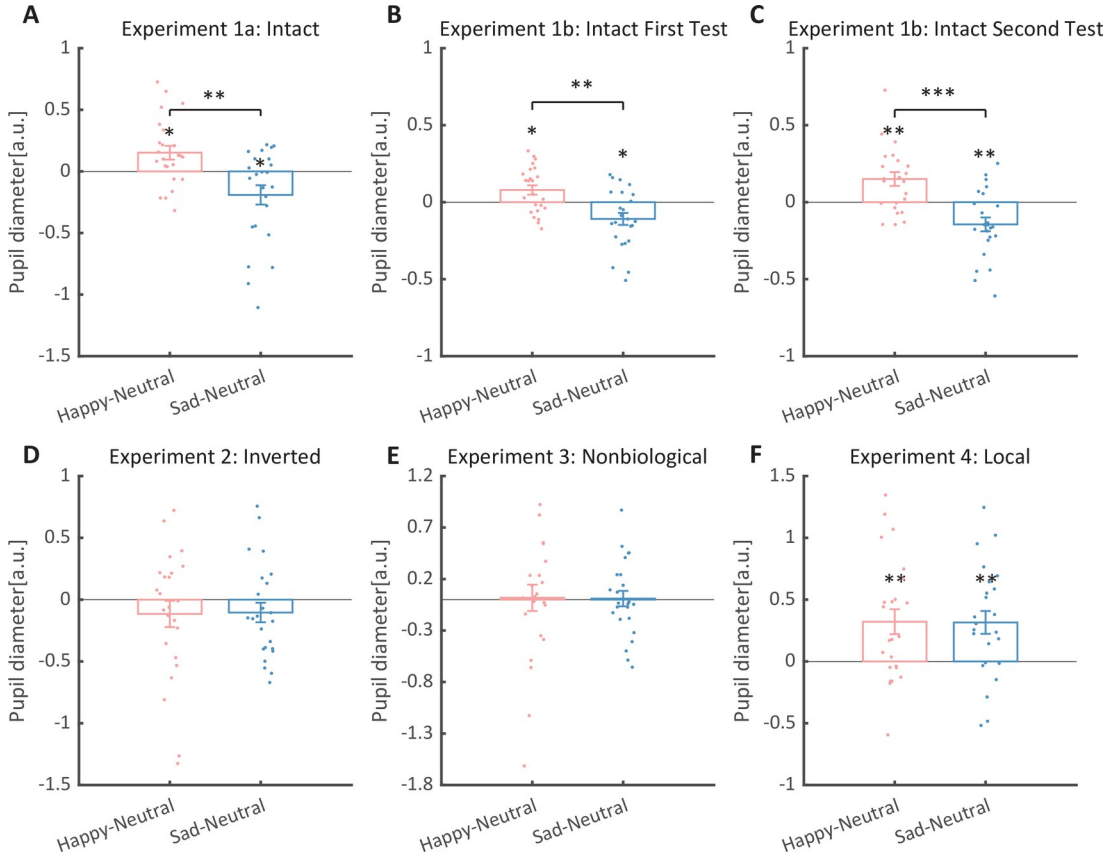
Yuan et al. (2024), Video 1.

Local motion and body form provide different information

The point-light study compared intact biological motion with displays preserving local movement while disrupting global body organization. **Local motion** describes the trajectories of individual points, whereas global form describes how those trajectories combine into a recognizable moving body. These manipulations separated two information sources present in the original stimulus.

Local motion still altered pupil responses, but it did not preserve the same fine distinction between happy and sad conditions as intact movement. It also produced an earlier, coarser modulation under the tested analysis. The physiological response therefore reflected several processing levels rather than only recognition of a complete human figure.

The intact display preserved relationships among moving joints, allowing the observer to extract both an overall body pattern and the trajectories of individual points. Disrupting that organization retained a faster, coarser pupil modulation while reducing the distinction between happy and sad motion. Global organization therefore contributed information beyond the presence of moving points alone.



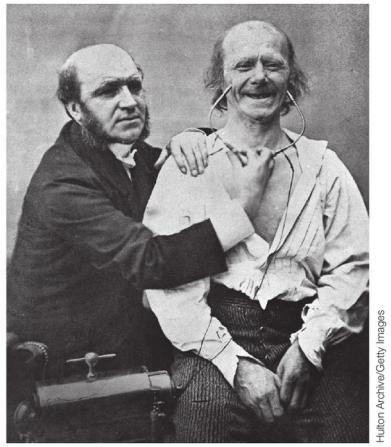
Yuan et al. (2024), Fig. 3.

Voluntary and spontaneous facial movements differ

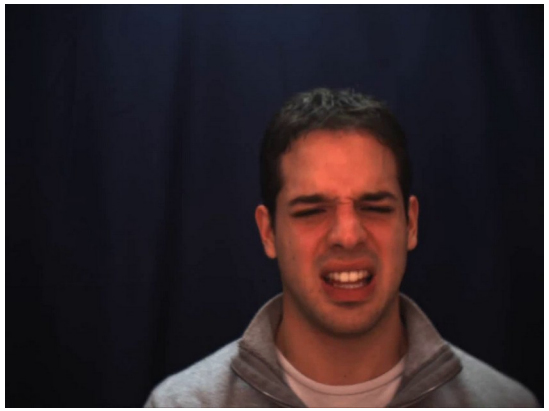
People can deliberately arrange their facial muscles, but spontaneous expression also emerges during an unfolding emotional response. **Voluntary facial movement** follows an intended action, while spontaneous expression is organized through partly different networks. The final muscles can overlap even though the routes recruiting them are not identical.

A deliberate smile can therefore differ in timing and coordination from a spontaneous smile. Movements around the eyes may contribute to the difference, but no single facial feature provides an infallible authenticity test. People can learn facial actions, and the meaning of a smile depends on the broader interaction.

Historical stimulation experiments showed that activating particular muscles could produce parts of an expression without generating the entire naturally coordinated pattern. These observations distinguish muscle contraction from the neural and social organization of an expressive act. The appearance of a facial movement alone does not establish the person's private emotional experience.



Carlson and Birkett (2022), Fig. 11.30, p. 366.



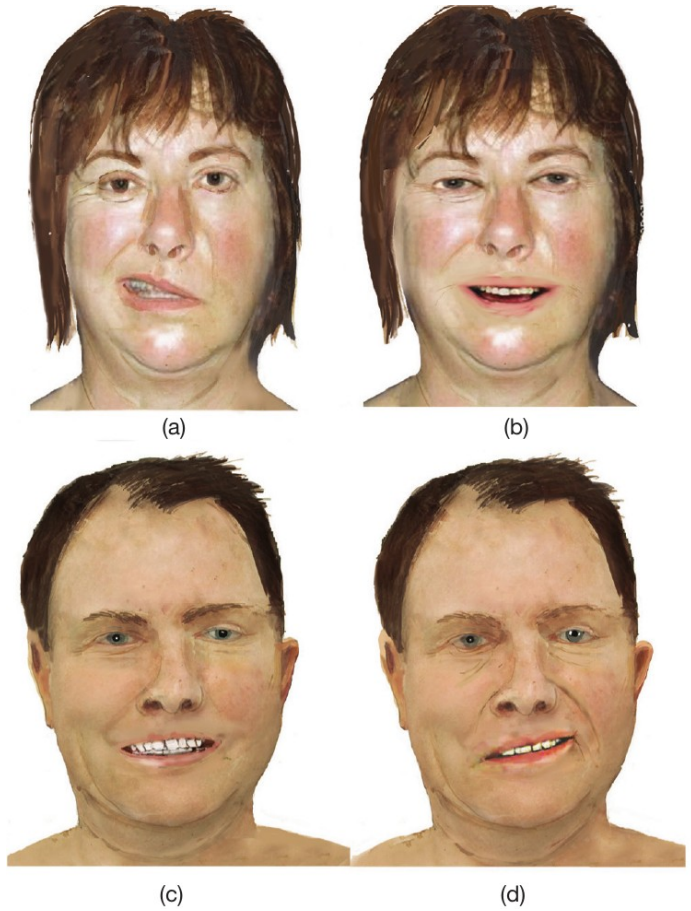
Jeganathan et al. (2022), Fig. 3-Video 9.

Lesions separate voluntary and emotional facial control

Volitional facial paresis is weakness in facial movement performed on command, while some spontaneous emotional movement remains relatively preserved. **Emotional facial paresis** shows the opposite kind of dissociation: commanded movement can remain more intact than the facial response emerging during an emotional situation. Paresis means partial weakness rather than complete paralysis.

These patterns indicate partially separable pathways that converge on the motor systems controlling facial muscles. A damaged route can impair one form of recruitment while another route still activates overlapping muscles. The difference cannot be explained simply by claiming that the muscle itself has lost every ability to contract.

Clinical comparisons require examining both sides of the face and several conditions, such as following a command and responding spontaneously. The lesion's location and extent also matter. The useful biological distinction is between routes of control, not a universal rule that all deliberate expressions are false or all spontaneous expressions reveal one unambiguous emotion.



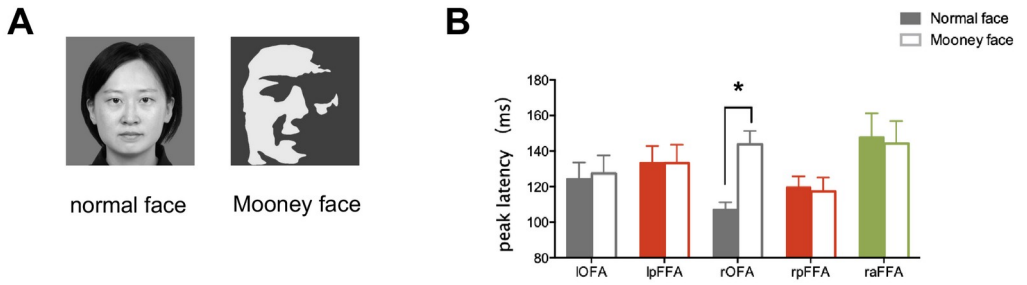
Carlson and Birkett (2022), Fig. 11.31, p. 367.

Face perception uses sensory detail and prior structure

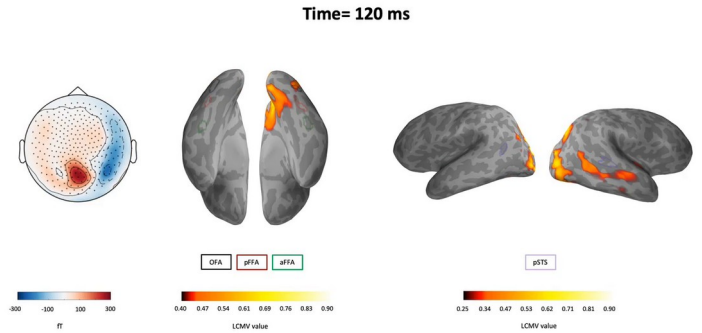
A human study combined fMRI with **magnetoencephalography**, or MEG, which measures magnetic fields associated with neural activity. The methods provided complementary anatomical and temporal information. Participants viewed ordinary faces and more ambiguous face-like displays, allowing the investigators to compare processing when familiar visual features were complete or difficult to extract.

Ordinary faces produced a sequence progressing from posterior visual regions toward more anterior face-processing areas. Ambiguous two-tone **Mooney faces** showed stronger evidence of top-down processing from fusiform toward occipital face-selective regions. Recognizing a face therefore did not always follow the same one-way sensory sequence.

Contextual cues could also recruit face-related responses when detailed internal features were absent. The study concerned mechanisms of face perception rather than a direct measurement of feeling. These perceptual operations provide information used in emotional communication, but detecting a face and interpreting its expression remain distinguishable stages of processing.



Fan et al. (2020), Fig. 3.



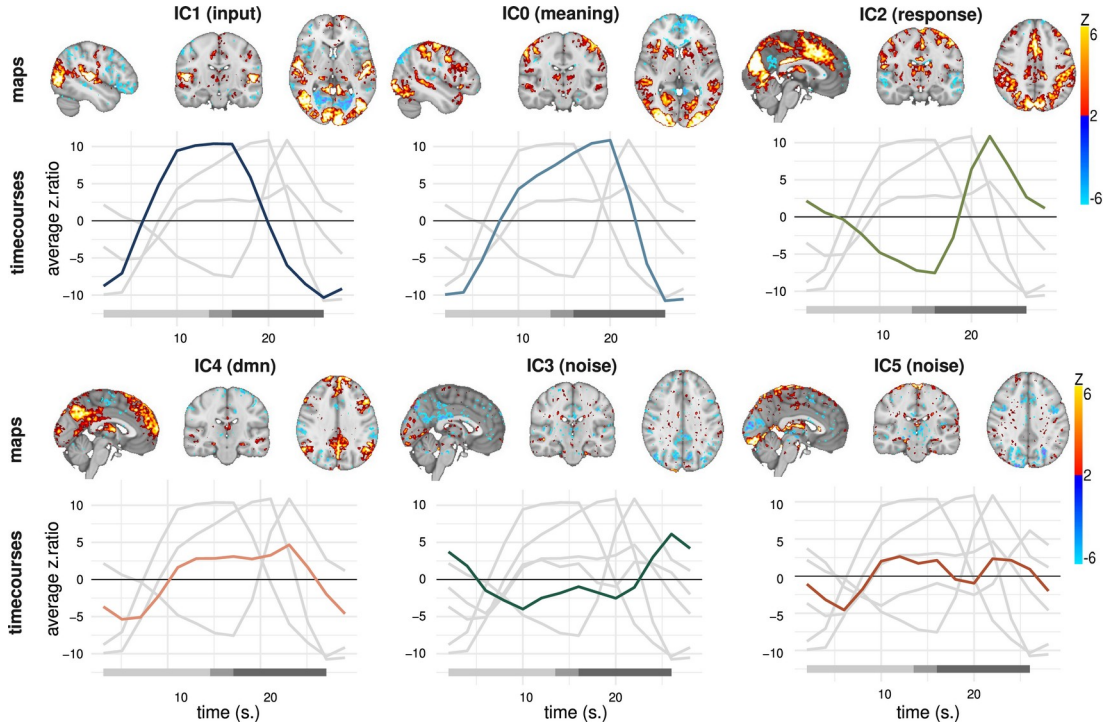
Fan et al. (2020), Video 1.

Human emotion recruits networks that unfold through time

A recent fMRI study examined responses while participants watched emotionally evocative movie clips and made emotion-related decisions. Rather than averaging the whole trial into one activation value, the researchers analyzed changing whole-brain patterns. This approach asked how perceptual evaluation, interpretation, and response-related processing followed one another during the task.

The analysis identified networks with different spatial distributions and time courses. Earlier patterns involved perceptual and affective evaluation, while later patterns were associated with semantic interpretation and decision demands. A further pattern aligned with default-mode activity. The sequence extended beyond one region that activated whenever any emotional clip appeared.

These networks were analysis-derived group patterns, not four anatomically isolated modules. Their timing also reflected the task structure and the slower blood-oxygen signal. The result showed how emotion-related processing could be examined as a changing distributed event rather than only a static map of regions averaged across several seconds.



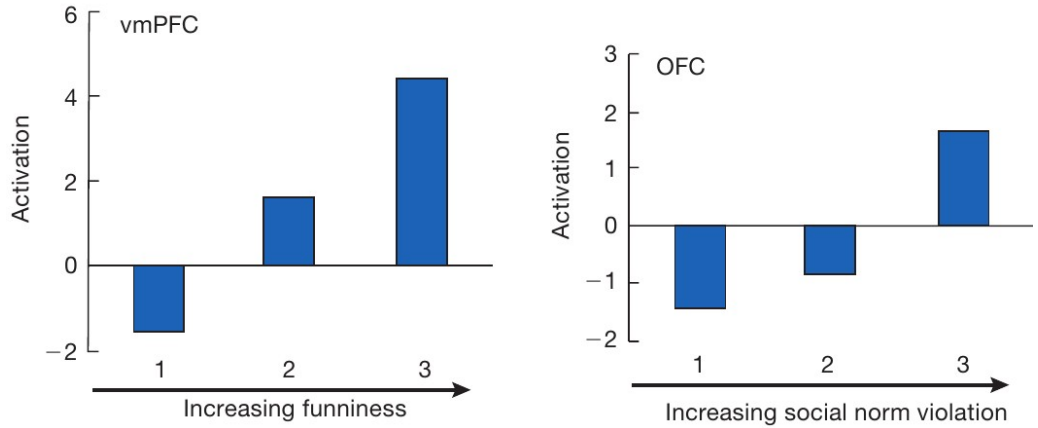
Janssen et al. (2026), Fig. 3.

Humor combines interpretation with a response

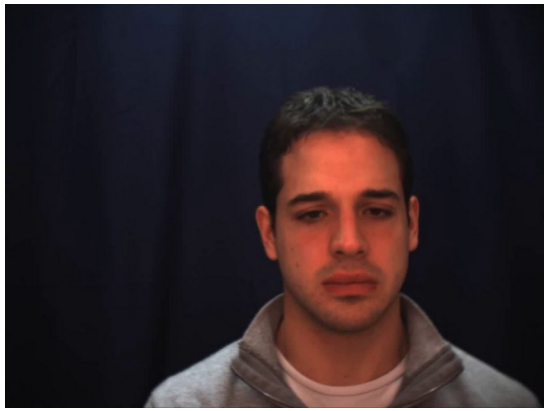
Humor often depends on detecting an unexpected interpretation, an incongruity, or a manageable violation of expectations. Recognizing the structure of a joke and enjoying it are related but different operations. A person can understand what was intended while finding it unfunny, unpleasant, or inappropriate in the current social setting.

Experiments implicate temporal, prefrontal, cingulate, and reward-related systems in different aspects of humor processing. Some comparisons separate perceived funniness from the degree of a social violation. The resulting activity cannot be reduced to one universal laughter center because comprehension, evaluation, and motor expression contribute different components.

Laughter can also occur during nervousness, affiliation, or unusual neurological conditions, while amusement need not produce visible laughter. Measuring a motor expression therefore does not establish the exact underlying experience. A mechanistic account distinguishes detecting the unexpected event, appraising its social significance, and generating the bodily or vocal response.



Carlson and Birkett (2022), Fig. 11.32, p. 368.



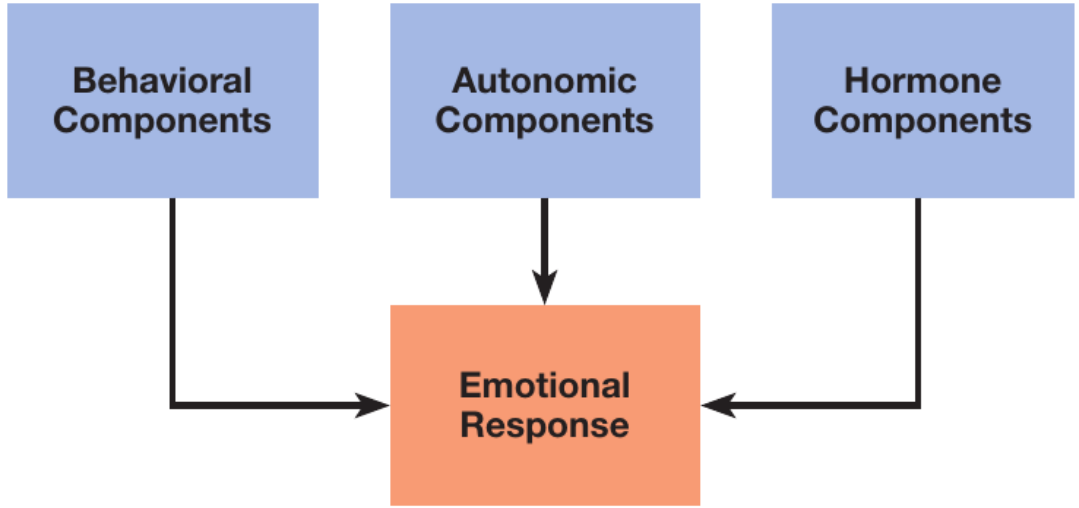
Jeganathan et al. (2022), Fig. 3-Video 10.

Feeling interacts with bodily and behavioral processes

Subjective experience is the felt aspect of emotion reported by a person. It is related to sensory interpretation, memory, bodily state, and ongoing action. A person can sometimes notice a racing heart before identifying the situation as frightening, or identify a danger before becoming aware of a strong bodily change.

Experimental reports depend on how a question is asked and when the participant answers. A momentary intensity rating and a later description of an entire event may capture different information. Neural and physiological recordings can complement these reports, but they do not remove the need to define the experience being measured.

An emotional episode also develops over time as new information arrives. Initial surprise may become relief, anger, or amusement when the cause is understood. The same bodily response can consequently be interpreted differently as knowledge and context change, making the sequence of processing important for understanding the final reported feeling.



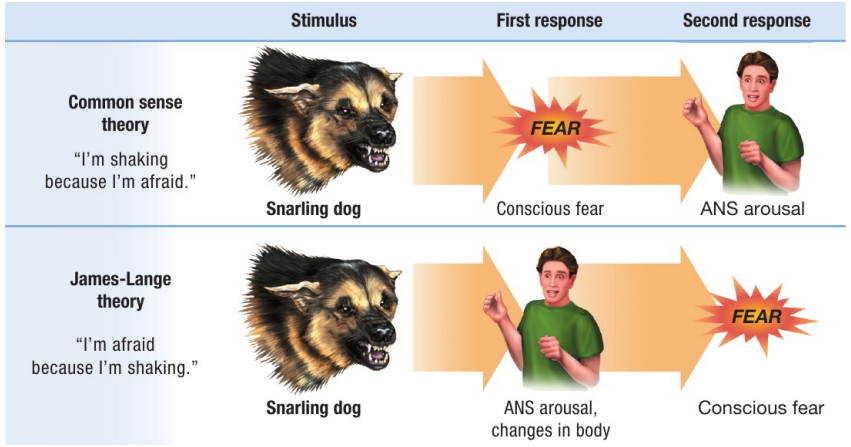
Carlson and Birkett (2022), Fig. 11.1, p. 339.

Classic theories ask how bodily feedback shapes feeling

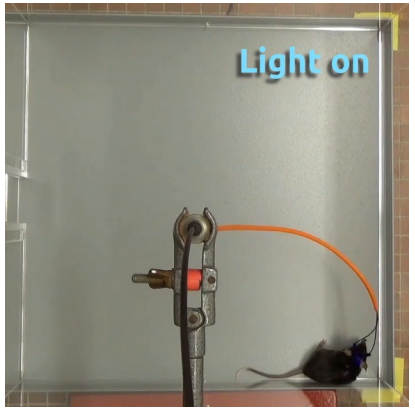
The **James–Lange account** proposed that perceiving changes in the body contributes to emotional feeling. On this view, an event triggers bodily and behavioral responses, and feedback from those responses helps constitute the experience. The proposal asks about information returning from the body rather than treating feeling as entirely separate from physiological action.

The **Cannon–Bard account** emphasized central processing that can organize bodily responses and feeling in parallel. One concern was that similar visceral changes occur across different emotions. A bodily pattern alone might therefore fail to specify whether a person is frightened, excited, angry, or engaged in strenuous physical activity.

Later accounts include cognitive appraisal, the interpretation of an event in relation to goals and context. These mechanisms need not be mutually exclusive in every respect. Contemporary experiments ask which bodily information is available, how it affects processing, and under what conditions it changes a reported feeling or decision.



Carlson and Birkett (2022), Fig. 11.33, first two theory rows, p. 370.



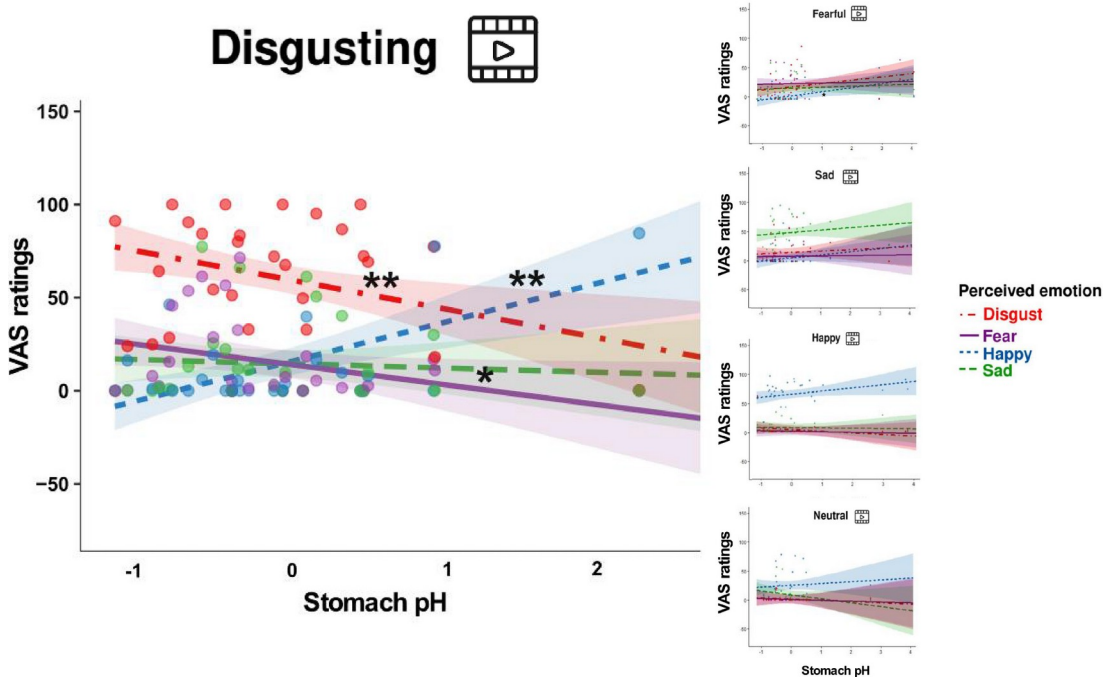
Cola et al. (2024), Video 1.

Swallowed sensors track the stomach during emotion

A human study used an **ingestible sensor**, a swallowed capsule measuring conditions inside the digestive tract. Participants viewed clips intended to evoke disgust, fear, happiness, sadness, or a neutral response. The investigators combined internal measurements with emotion reports, visceral sensations, and cardiac measures rather than inferring stomach physiology solely from a questionnaire.

During disgust-related clips, more acidic stomach conditions were associated with stronger reported disgust and fear, while less acidic conditions were associated with greater happiness in the tested analysis. **pH** measures acidity: lower pH indicates a more acidic environment. This association linked a directly measured internal condition with reported experience.

The experiment observed physiological variation during emotion induction rather than independently imposing stomach acidity as the cause of a feeling. It therefore did not establish a simple pathway in which lowering pH necessarily produces disgust in any person. The sensor supplied access to a bodily variable that conventional surface measures could miss.



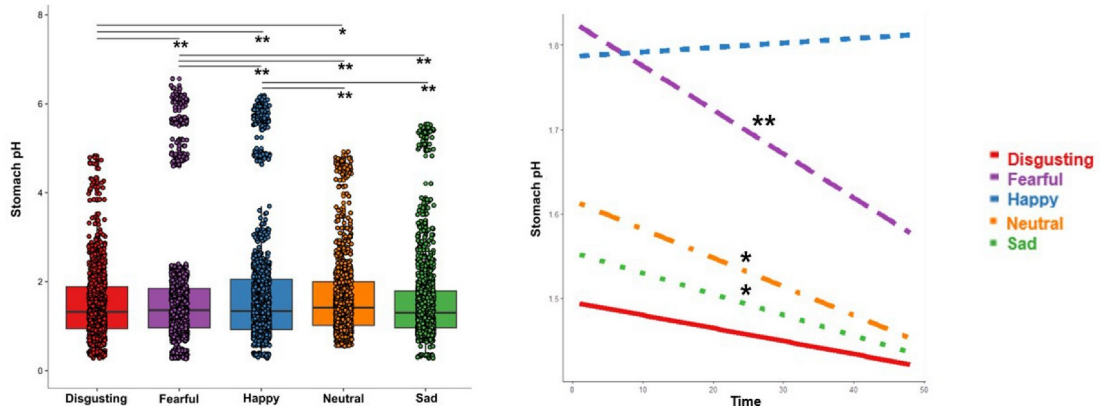
Porciello et al. (2024), Fig. 5.

Different bodily measures do not all track the same emotion

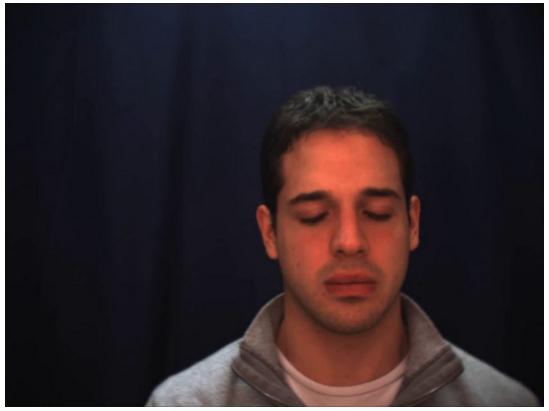
The ingestible-sensor study measured pressure and temperature along with pH and collected reports of gastric, respiratory, and cardiac sensations. **Visceral sensations** arise from internal organs, such as nausea or awareness of heartbeat. Comparing these measurements tested whether a reported emotional state corresponded to one uniform change throughout the body.

Fearful and disgusting clips were accompanied by particular patterns of reported bodily sensation and cardiac response. The stomach-related associations were not identical to every measure from other digestive locations or every emotional condition. A bodily contribution to emotion therefore needed to be specified by organ, variable, and the stimulus being viewed.

The capsule traveled through the stomach and later intestinal regions, allowing the investigators to compare internal measurements at different digestive locations. The pH, pressure, and temperature streams described local physiological conditions, while reports described the participant's experience. Comparing those locations prevented a stomach-related association from being treated as the same response everywhere in the digestive tract.



Porciello et al. (2024), Fig. 6.



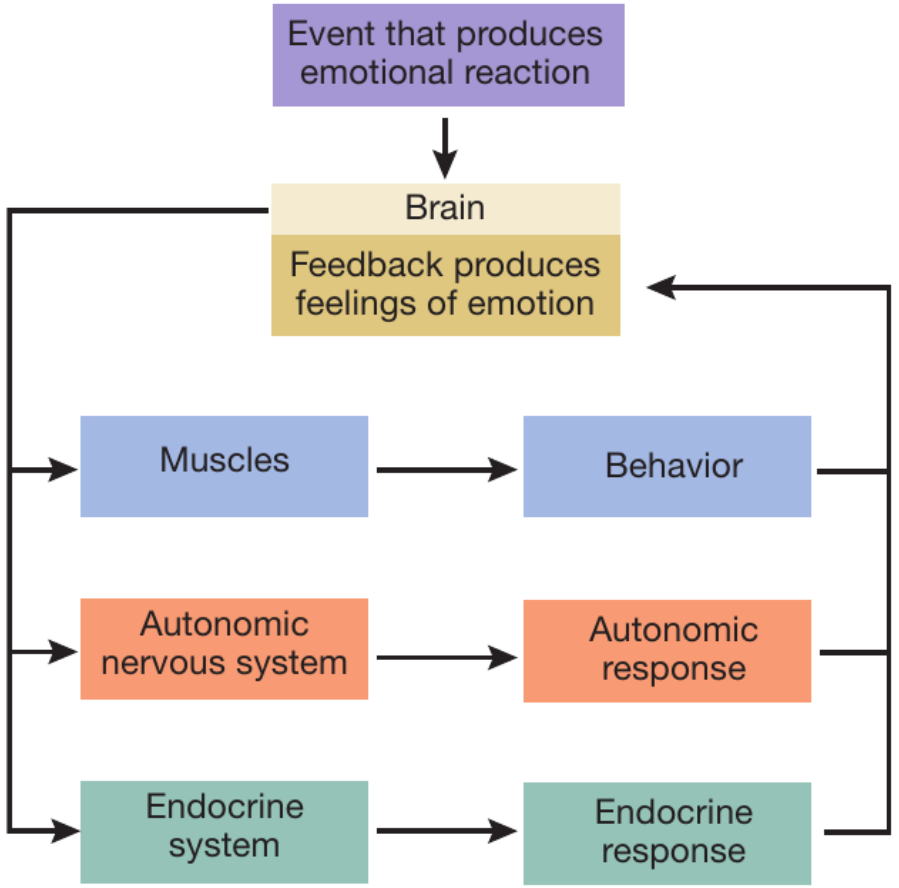
Jeganathan et al. (2022), Fig. 3-Video 11.

Interoception carries information from inside the body

Interoception is the processing of signals about the body's internal condition, including cardiovascular, respiratory, and digestive activity. Sensory pathways travel through peripheral nerves, spinal routes, and brainstem relays toward broader networks. Much of this information regulates physiology without necessarily becoming a conscious sensation at every moment.

The **vagus nerve** carries important visceral sensory information as well as parasympathetic motor signals, but it is not the only route for bodily feedback. Brainstem, thalamic, insular, and cingulate systems help combine internal signals with current demands. Neural regulation and conscious awareness therefore overlap without being identical processes.

A prediction about the body can also influence how an incoming signal is interpreted. Feeling a heartbeat during exercise differs from noticing the same sensation in an uncertain medical situation. Interoception contributes to experience through an interaction between actual bodily input, prior knowledge, attention, and the meaning assigned to the event.



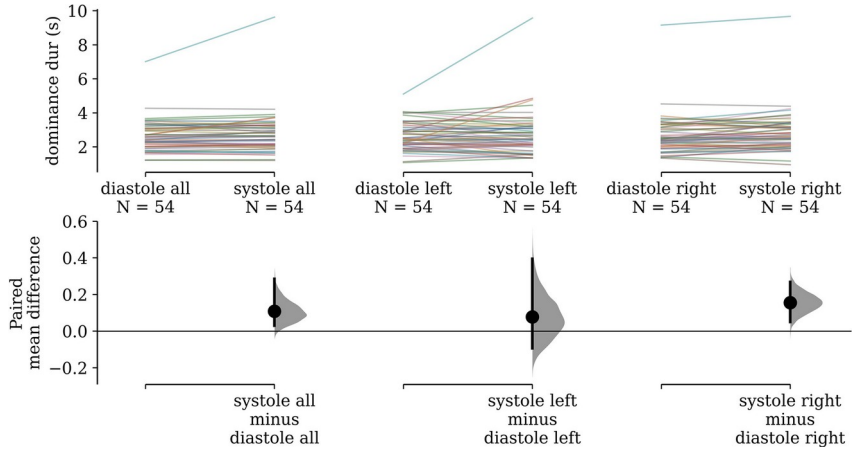
Carlson and Birkett (2022), Fig. 11.34, p. 371.

Cardiac timing can alter which visual stimulus dominates

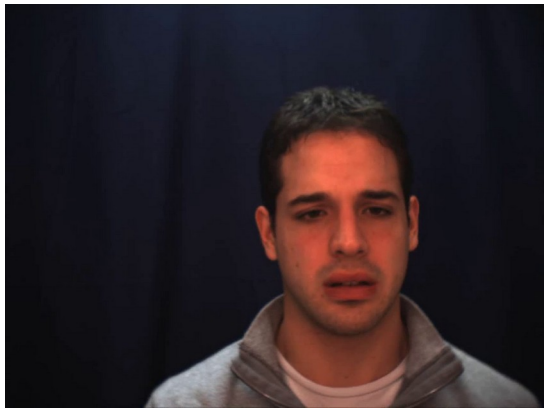
A human experiment presented different grating patterns to the two eyes, producing **binocular rivalry**, in which perceived dominance alternates between competing images. The patterns pulsed at particular phases of the participant's heartbeat. This tested whether internal cardiovascular timing could influence visual awareness under otherwise similar sensory conditions.

During **systole**, heart contraction activates pressure-sensitive **baroreceptors** that signal toward the brain. A comparison with **diastole**, the relaxation phase, allowed the investigators to relate stimulus dominance to the timing of this bodily input. The study manipulated when the visual pulse occurred, not the participant's emotional interpretation of a threatening picture.

Patterns synchronized to systole remained perceptually dominant for longer under the tested conditions. The result opposed a simple claim that cardiac afferent input always suppresses vision. A bodily signal could influence sensory competition in a way that depended on the task and temporal relationship between internal and external events.



Veillette et al. (2024), Fig. 2.



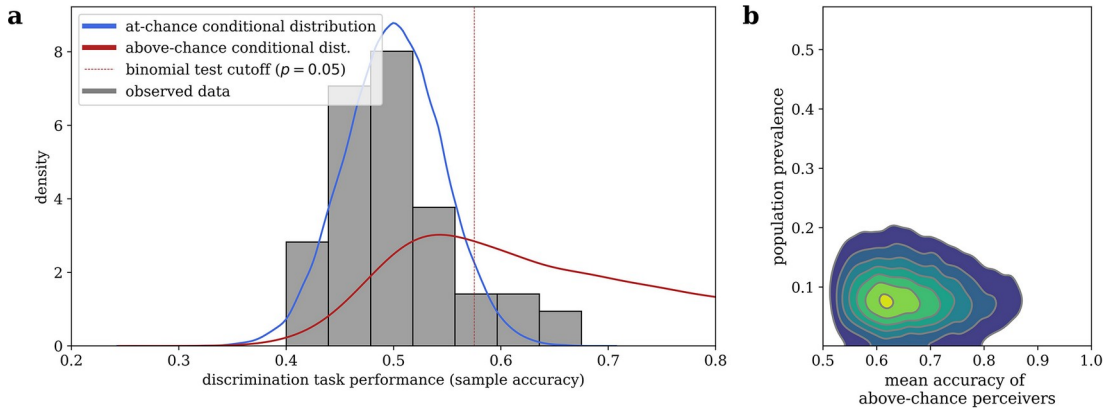
Jeganathan et al. (2022), Fig. 3-Video 12.

Bodily influence can occur without conscious recognition

The cardiac-rivalry study separately assessed whether participants could discriminate the relationship between stimulus timing and their heartbeat. **Interoceptive awareness** concerns conscious access to an internal signal and is not identical to the signal's physiological arrival in the nervous system. Someone can process bodily input without reporting its timing accurately.

The visual-dominance effect persisted in participants whose timing judgments were near chance in the separate task. Their cardiac-linked sensory modulation therefore did not depend on successful conscious identification of each heartbeat relationship. The effect concerned how internal timing influenced perception rather than only what participants believed they were detecting.

The analysis modeled individual variation in discrimination rather than assuming all participants had identical bodily awareness. This separated a physiological contribution to visual processing from explicit performance on a heartbeat-related judgment. Both matter for theories of bodily experience, but one cannot simply be used as a substitute measure for the other.



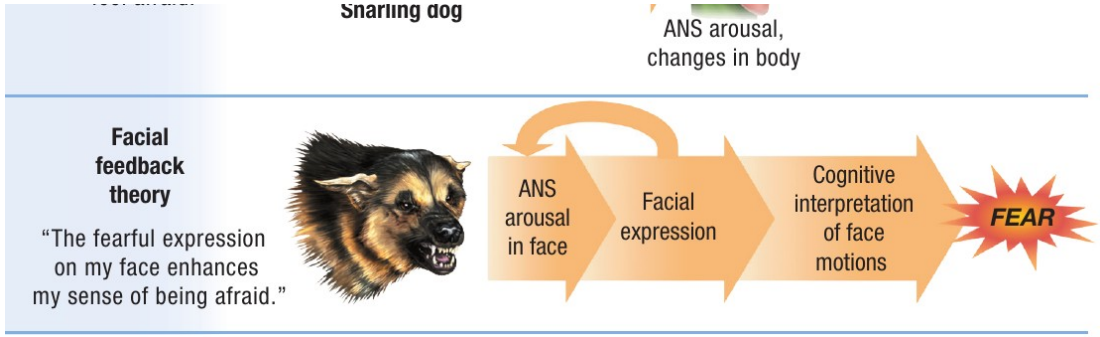
Veillette et al. (2024), Fig. 3.

Facial feedback can modulate emotional experience

The **facial-feedback hypothesis** proposes that sensory information from facial action can influence emotional processing or experience. Moving a muscle changes signals from the face as well as what other people see. The hypothesis concerns feedback from the expressive action rather than a claim that a smile guarantees happiness in every setting.

Experiments manipulate facial posture or movement and then measure reports, judgments, or physiological responses. Results depend on the procedure, context, and outcome, and effects are not equally strong across all tasks. Simply asking someone to arrange the mouth differently does not isolate every possible sensory and cognitive consequence of that instruction.

Expectations and interpretation can contribute alongside facial sensory signals. A participant who knows the intended emotion may respond differently from one performing a less obvious motor task. Evaluating facial feedback therefore requires distinguishing the muscle action, the information it returns, and the participant's understanding of what the experiment is asking them to do.



Carlson and Birkett (2022), Fig. 11.33, facial feedback row, p. 370.



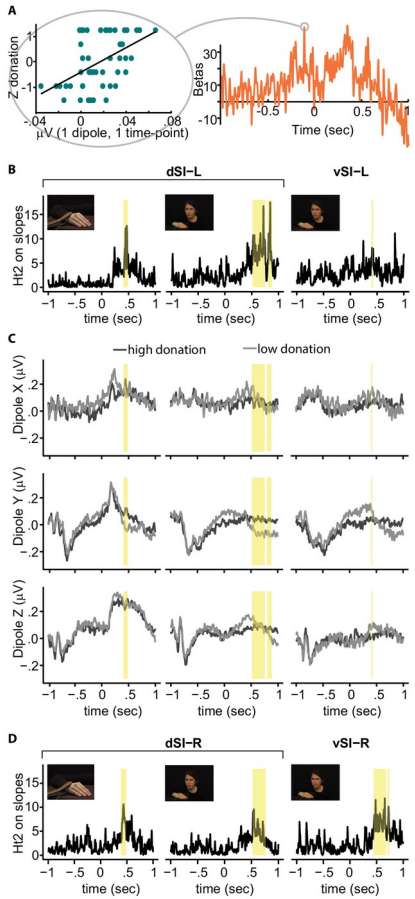
Jeganathan et al. (2022), Fig. 3-Video 14.

Observed bodily reactions can influence a decision to help

A human study showed participants videos of another person responding to painful stimulation through either hand movement or facial expression. Participants could donate money to reduce the other person's pain. The design linked a controlled sensory cue to both perceived pain and a concrete prosocial choice rather than measuring empathy only through a general questionnaire.

Electroencephalographic activity in the hand area of **primary somatosensory cortex**, or SI, explained variation in donations under the tested conditions. This cortex normally processes bodily sensory information, but observed hand responses could also recruit it. The analysis asked whether this vicarious activity contributed information used in the subsequent helping decision.

The source videos varied the intensity of visible reactions, allowing response strength to be compared across conditions. A donation is a behavioral outcome and does not directly measure every aspect of felt compassion. The experiment could therefore test the sensory-to-choice transformation while keeping the observed action separate from an inferred private emotional state.



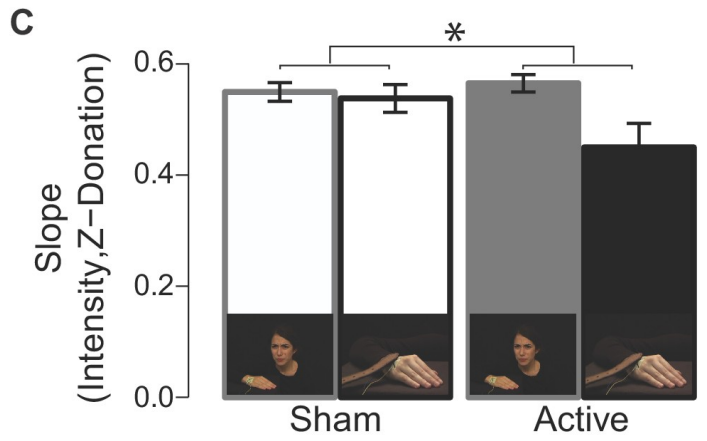
Gallo et al. (2018), Fig. 3.

Somatosensory disruption affects hand-cued helping

The observed-pain study used **transcranial magnetic stimulation**, or TMS, to alter activity in the somatosensory hand region. This method applies a magnetic pulse that changes cortical activity near the target. It tested whether a region associated with donations also made a causal contribution to using the observed bodily information.

TMS disrupted the coupling between observed pain intensity and donations when pain was expressed through the hand, but not in the same way when it was expressed through the face. The effect was therefore tied to the bodily information being represented rather than a uniform inability to make any helping decision.

Additional electrical-stimulation experiments examined perceived pain judgments. Together, the interventions helped separate sensory representation from the final social choice. A somatosensory contribution can support understanding another person's bodily response without implying that the observer literally experiences the identical pain or that this region alone produces altruism.



Gallo et al. (2018), Fig. 2.



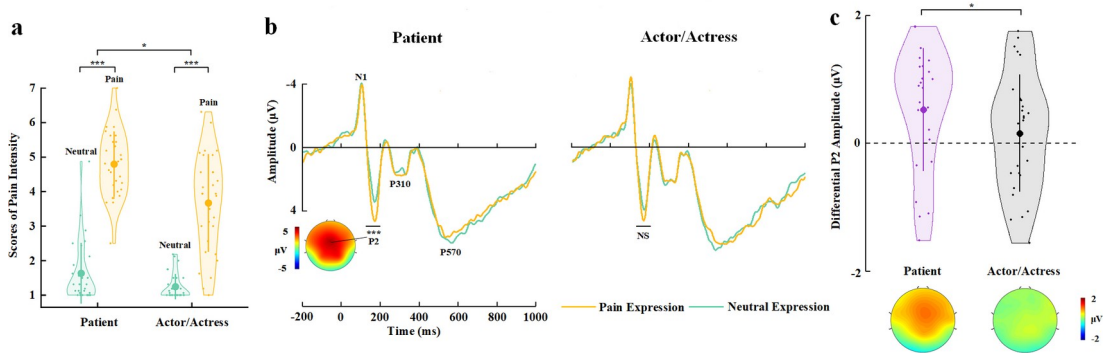
Gallo et al. (2018), Video 2.

Beliefs about another person change responses to their pain

A human study changed what participants believed about another person's painful expression. The visible cue was not the only information available: participants also learned whether the person was thought to be genuinely experiencing pain. **A belief about another's pain** can therefore alter the meaning assigned to an otherwise similar sensory display.

Reduced belief in the other person's pain lowered estimated pain and monetary donations. Neural measurements also showed changes in early responses and in insular, somatosensory, and frontal regions. The social interpretation influenced processing at several levels rather than merely changing a final verbal answer after identical internal responses.

The experiment distinguished the physical expression from the participant's knowledge about its source. This is essential when interpreting an empathy-related brain response. Activity can reflect attention, salience, belief, and affective sharing in different combinations, so a region activated by another person's pain does not uniquely establish one psychological process without the relevant comparison.



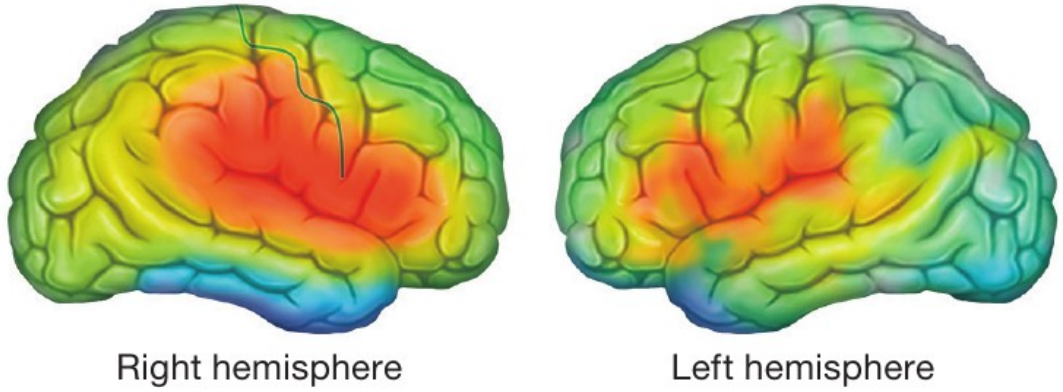
Wu et al. (2021), Fig. 3.

Empathy, compassion, and helping are related but distinct

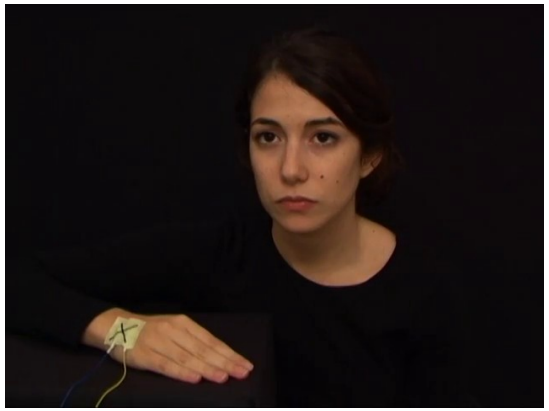
Empathy can include sharing or understanding another person's state, while **compassion** involves concern for their welfare. **Prosocial behavior** is an action that benefits another individual, such as providing assistance. These processes can interact, but feeling distressed, understanding a need, and choosing to help are not interchangeable measurements.

An observer may understand someone's situation without sharing its full emotional intensity, or share distress while becoming too overwhelmed to act effectively. Helping can also reflect social rules, learned habits, and deliberate commitments. A measured action therefore needs its own explanation rather than being assigned automatically to one presumed feeling.

Experiments compare cues, beliefs, costs, and opportunities to identify what changes the choice. Lesion and stimulation studies can examine particular contributions within that sequence. The biological account is strongest when it identifies how information about another person affects the action, rather than naming one general empathy network as the complete mechanism.



Carlson and Birkett (2022), Fig. 11.28, p. 364.



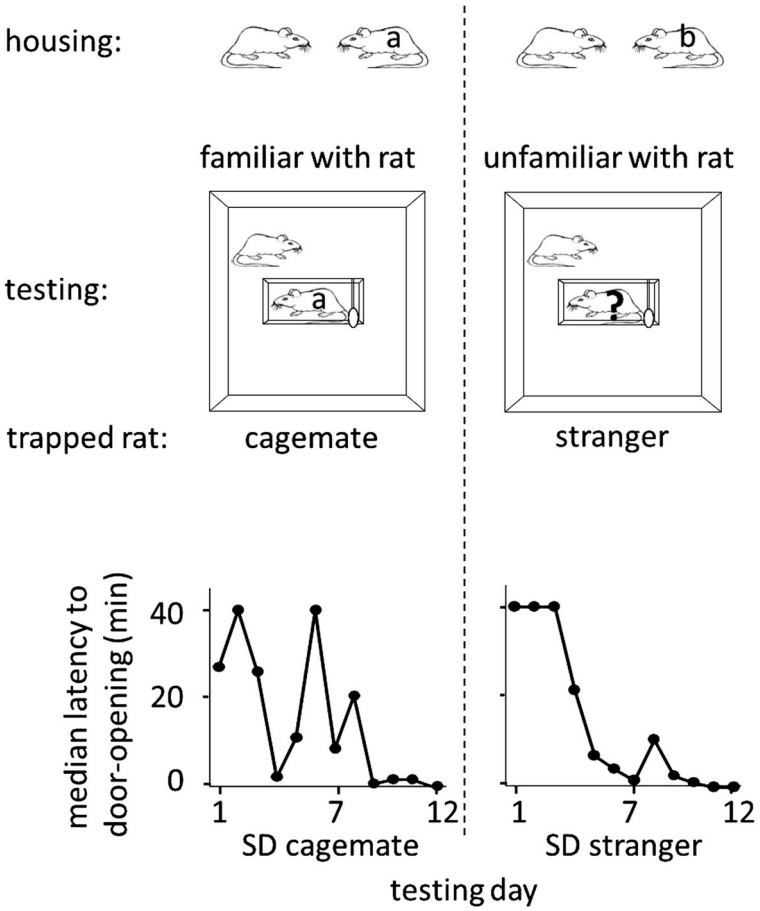
Gallo et al. (2018), Video 4.

Experience changes which unfamiliar rats receive help

A rat study examined opening a restrainer containing another rat. **A restrainer** limits the trapped animal's movement while allowing the free animal to interact with its door. The experiment compared familiar and unfamiliar strains, asking whether prior social experience influenced helping toward an individual the subject had never previously encountered.

Rats helped unfamiliar animals of a familiar strain more readily than unfamiliar animals of a strain they had not experienced. Housing with even one member of another strain changed responses toward additional animals of that strain. Experience with one individual could therefore generalize to the behavioral treatment of other, similar-looking conspecifics.

Door opening was the measured outcome, and social contact and learning were important features of the task. The finding did not require assuming a human-like verbal concept of group membership. It showed that an observable prosocial response could be reorganized by social familiarity rather than being fixed exclusively by genetic similarity.



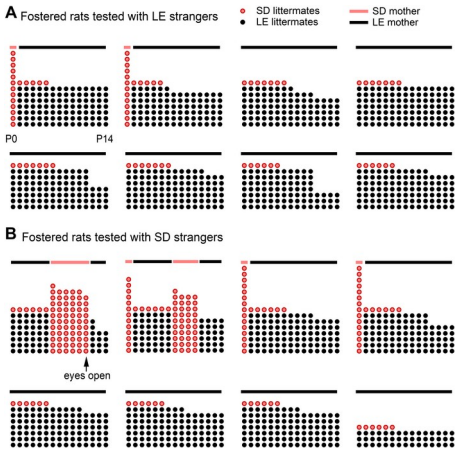
Ben-Ami Bartal et al. (2014), Fig. 1.

Cross-fostering separates genetics from social familiarity

The rat helping study used **cross-fostering**, raising young animals with members of a different strain. This changed the social group experienced during development without changing the offspring's own genetic strain. The design tested whether inherited similarity alone predicted which unfamiliar trapped animals would later receive help.

Cross-fostered rats helped strangers of the strain they had been raised with more readily than strangers of their own unfamiliar strain. Genetic matching was therefore insufficient to explain the response. The subject's social experience determined which category of conspecifics had become familiar enough to recruit the measured helping action.

The cross-fostered subjects had developed without ordinary contact with members of their own strain. Their later responses therefore separated physical similarity to themselves from social familiarity acquired during rearing. Helping generalized toward the experienced strain even when the trapped animal was a new individual, showing that the effect extended beyond recognition of one particular caregiver or companion.



Ben-Ami Bartal et al. (2014), Fig. 3.



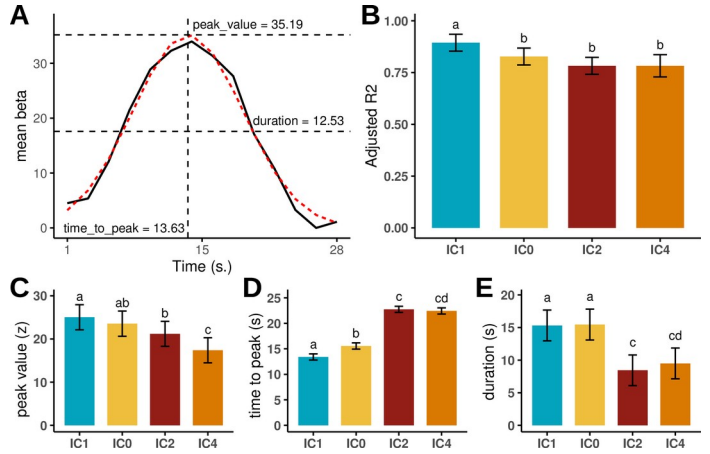
Ben-Ami Bartal et al. (2014), media2.

Emotional content changes distributed processing over time

The dynamic-fMRI study compared the peak, duration, and relative timing of network responses to emotionally different movie clips. These features contain information lost when a whole trial is reduced to one average. A network can respond for longer in one condition even when its maximum signal is not the largest.

Happy, sad, and fearful content modulated spatial and temporal properties across the identified networks. The effect depended on the network and the measured feature rather than following one uniform ordering across the entire brain. Perceptual evaluation, interpretation, decisions, and later task-related activity unfolded on overlapping but distinguishable time courses.

The blood-oxygen response is slower than the underlying neural events, so its timing must be interpreted through the imaging method and experimental design. Even with that constraint, time-resolved analysis distinguished stages that a static map concealed. The result identifies changing coordination across distributed systems as part of how an emotional task develops through time.



Janssen et al. (2026), Fig. 7.



Ben-Ami Bartal et al. (2014), media1.

Key takeaways

Emotion has several measurable components. Behavior, autonomic and endocrine responses, and subjective feeling can vary together or separately; one physiological measure does not identify every emotional state.

Threat responses emerge from interacting circuits. Amygdala, hypothalamus, brainstem, and cortical pathways shape learning and select among freezing, escape, and other defensive actions.

Learning changes both responses and their context. Extinction creates new learning, while counterconditioning adds an opposing outcome; persistence and return of defensive responses depend on the tested conditions.

Aggression and control depend on specific pathways. Hormones, serotonin, experience, and prefrontal–subcortical circuitry influence distinct actions; no single transmitter or region explains all human violence.

Emotional communication unfolds through time. Facial action, gaze, posture, voice, and context contribute to recognition; observed expression is evidence about a response rather than direct access to private feeling.

Bodily feedback and social knowledge shape experience. Interoceptive signals contribute to processing, while beliefs and social experience alter empathy and helping; these processes require separate behavioral and neural measurements.